XAnnoRank Documentation

XAnnorank is an extension of AnnoRank that supports explainability studies in rankings via user-centered evaluation. XAnnorank provides an easily adaptable user interface (UI) that enables systematic assessment of explainability methods by presenting ranked lists with and without explanations, while collecting both implicit and explicit user feedback. Its flexibility facilitates diverse user design, including different explanation types, datasets, ranking models, and researcher-defined configurations.This updated version supports as well the previous functionalities of AnnoRank: collect interactions between the user and the ranked list of items, collect graded relevance for an item given the displayed query, and compare two rankings and assess which ranking is more suitable given the query and the assessment's requirements. Moreover, AnnoRank offers the researcher the possibility to view the annotations collected and compare two rankings as well as viewing the corresponding evaluation metrics.

Summary of updates:

- Display of explanations:
    - simple text
    - image
    - text and image
    - factual
    - counterfactual
- Built-in questionnaire
- Improved attention check
- Collection of implicit and explicit evaluation signals for the explanations
- A recruitment use case

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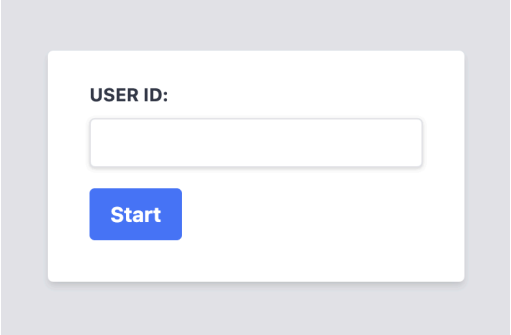
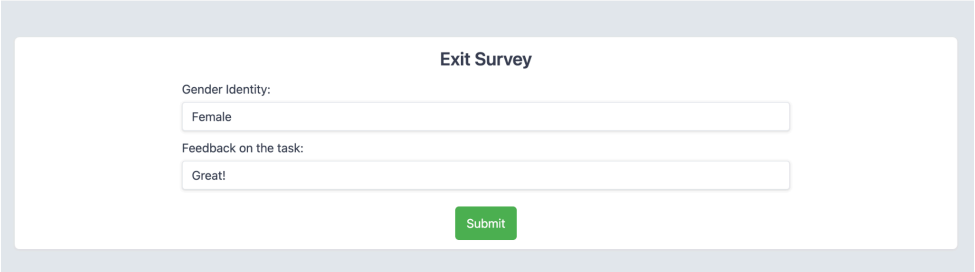
1. UI Functionalities

1.1 Collection of User Data

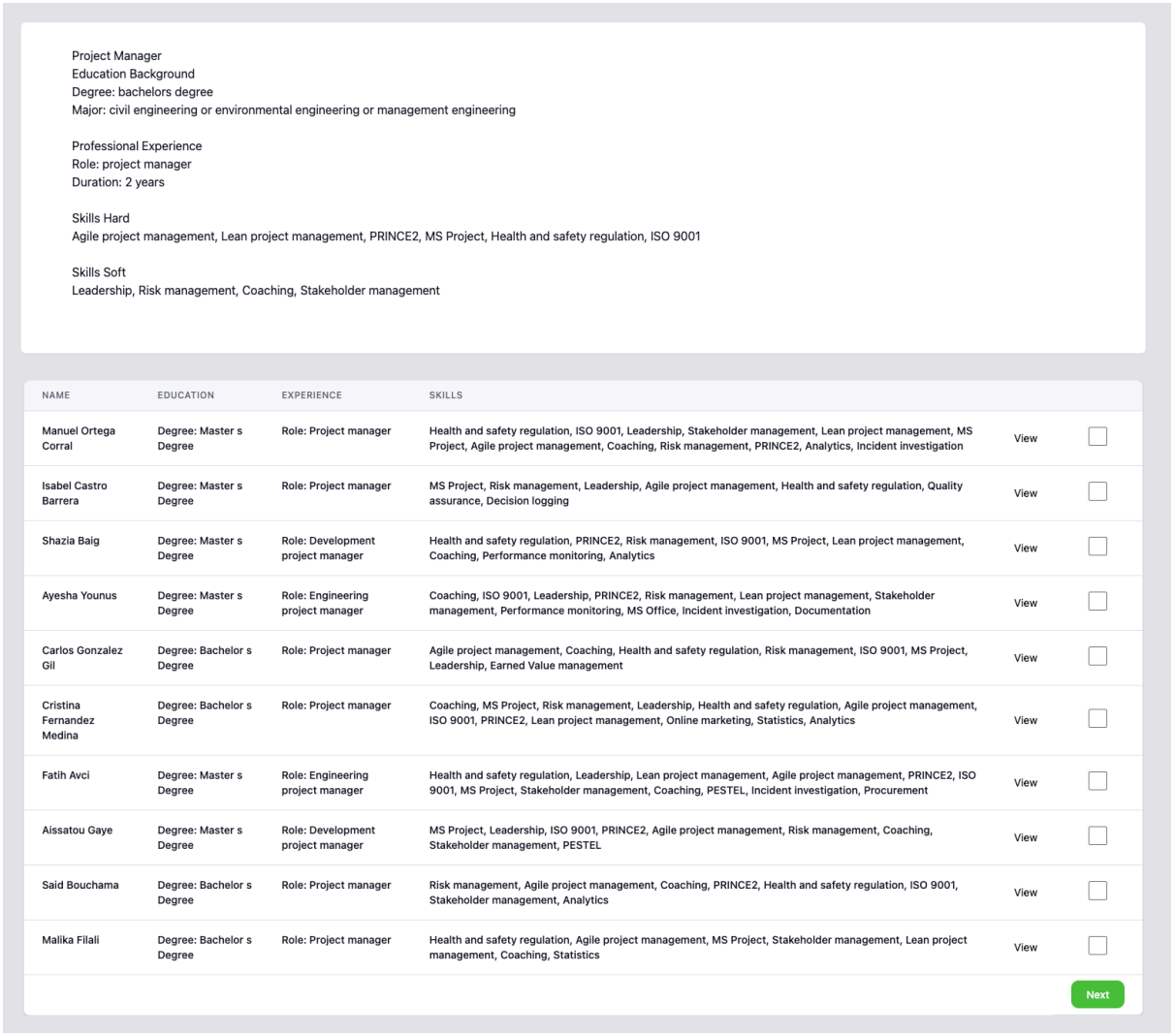
When interacting with the tool the users will be first asked to provide an ID with which they will be registered in the database. The user can interact with the UI by typing in the empty field the ID and then clicking start. All interactions will be collected during the session for each user ID.



At the end of the interaction with the tool, the users can be asked to complete an exit survey in which one can collect feedback or demographic data about the user. An example of an exit survey can be seen below. This can be configured to include other fields. More details about the configuration of the Exit Survey can be found under the section Configuration File. The user can interact with the UI by typing in the empty fields (e.g. feedback field) or by choosing an option from a drop-down (e.g. gender field), then when clicking submit all the collected values are stored in the database for the corresponding user ID.

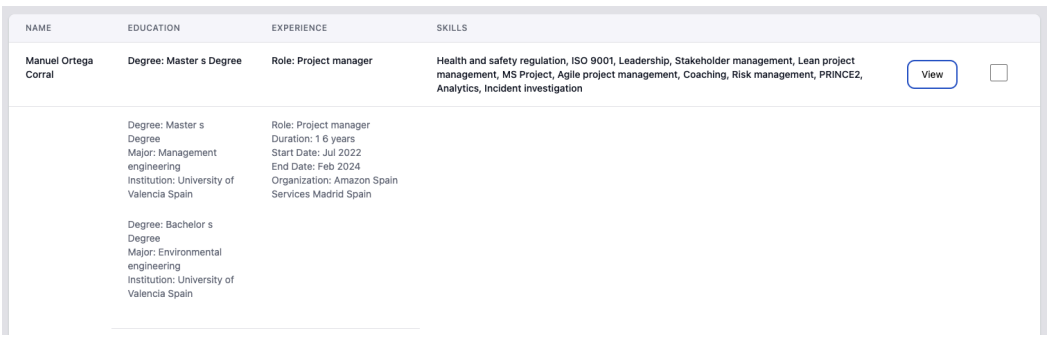


1.2 Display Interaction Annotate UI



In the option for displaying a query-ranking pair the UI will show in the top of the page the query, and in the bottom of the page the list of documents. The user can interact with the UI in the following ways:

> **View Button:** the user can expand the table to view more information about the document. When the user clicks on another View Button, the current expand will close. Only, one expand can be open at a time.



Check Box: the user can select top-k documents that are the most relevant given the displayed query. If the predefined minimum and maximum of selected documents is not met, the UI will display an alert to the user.

Next Button: the user can navigate to the next query using the Next Button.

For each user, the following interactions can be collected:

- **Clicks:** number of clicks for each document on the View Button
- **Timestamps:** for each click the timestamp of the click to open the expand and the timestamp of closing the expand are collected.
- **Top-k Documents:** the list of the selected documents in the order in which the user checked the box

Integration with BigBrother Logging Tool

On top of the above mentioned, AnnoRank is integrated with the BigBrother logging tool. The BigBrother logging tool is one of the application-independent services, with minimal configuration to the web-based user interface [8] (Github source: https://github.com/hscells/bigbro).

The tool can record the **mouse clicks** and **movements** of the annotators. The logs are stored in the ./dataset/<data_name>/*big_borther.log* file.

To collect *mouse clicks* insert this script into the html body:

```
<script type="text/javascript">
    BigBro.init('**{{session_id}}**', "localhost:1984", ["click",
"onload"]);
</script>
```

To collect *mouse movements* insert this script into the html body:

```
<script type="text/javascript">
    BigBro.init('**{{session_id}}**', "localhost:1984", ["mousemove",
"onload"]);
</script>
```

The **session_id** is the `"user_id"` of the current annotator.

Below is an example of the Big Brother logging tool output for *clicks* when used in the Interaction Annotation UI app:

```
2024-06-03 09:50:26.303 +0000
UTC,**3456**,click,INPUT,shortlist,1,http://localhost:5000/start_ranking/1/in
dex_ranking/0/view,1229,423,1360,807,
2024-06-03 09:50:26.693 +0000
UTC,**3456**,click,INPUT,shortlist,2,http://localhost:5000/start_ranking/1/in
dex_ranking/0/view,1233,508,1360,807,
2024-06-03 09:50:27.14 +0000
UTC,**3456**,click,TD,,,http://localhost:5000/start_ranking/1/index_ranking/0
/view,1239,587,1360,807,
```

```
2024-06-03 09:50:27.477 +0000
UTC,**3456**,click,INPUT,shortlist,3,http://localhost:5000/start_ranking/1/index_ranking/0/view,1236,596,1360,807
```

Below is an example of the BigBrother logging tool output for the *mousemove* in score Annotation UI:

```
2024-06-03 11:19:25.017 +0000
UTC,**23445**,mousemove,H4,,,http://localhost:5003/start_annotate/2/index_annotate/1,692,450,1360,807,
2024-06-03 11:19:25.033 +0000
UTC,**23445**,mousemove,H4,,,http://localhost:5003/start_annotate/2/index_annotate/1,949,368,1360,807,
2024-06-03 11:19:25.05 +0000
UTC,**23445**,mousemove,DIV,,,http://localhost:5003/start_annotate/2/index_annotate/1,1129,336,1360,807,
2024-06-03 11:19:25.069 +0000
UTC,**23445**,mousemove,DIV,,,http://localhost:5003/start_annotate/2/index_annotate/1,1303,322,1360,807,
```

The logs follow the structure:

`<Time of logged event>, <User identifier> , <Method causing the event>, <HTML element>, <Name of HTML element> , <ID of HTML element> , <URL> ,<X position> ,<Y position> , <ScreenWidth> , <ScreenHeight> , <Comment>`

1.3 Score Annotate UI

In the option for annotating a document by giving it a score, the UI will show in the top of the page the query, and in the bottom of the page document. The user can interact with the UI in the following ways:

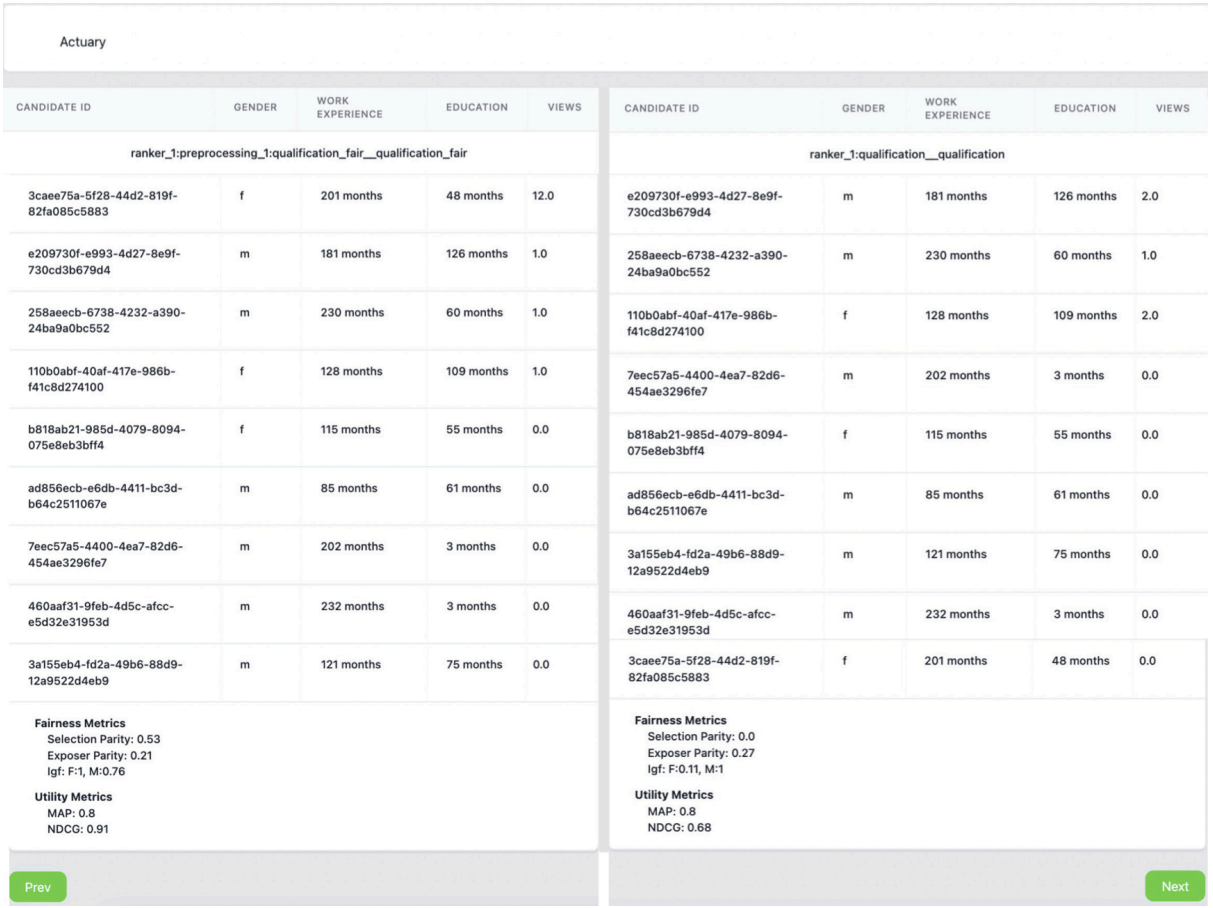
- **Score Bar:** the user can give a score to the document by clicking on the numbers in the score bar.
- **Next Button:** the user can navigate to the next query using the Next Button.

For each user, the following interactions can be collected:

- **Score:** the score the user things the document should receive given the displayed query and the task description.

1.4 Display Ranking Comparison UI

Ranking Compare Visualise UI

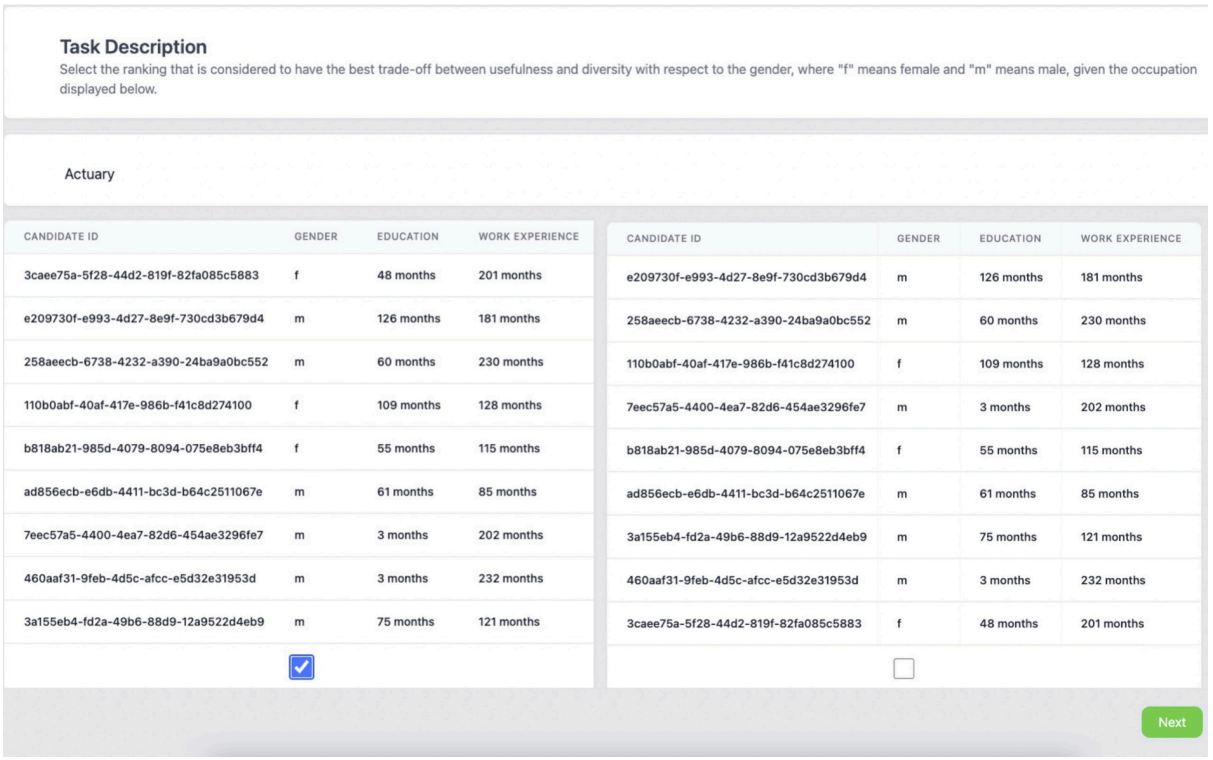


On top of this, this UI can be used by the researcher to compare rankings produced using different ranking methods or evaluate the impact of a fairness intervention. The UI has the option to display metrics for each ranking, as shown in the bottom of the UI. Moreover, if interaction data is available, the UI can display the average of the interactions collected by the users. The navigation between several ranking comparison can be performed using the **Prev/Next Button**.

Evaluation Metrics supported by AnnoRank

The utility metrics that can be displayed are computed using the Pytrec eval library[4] on the displayed ranking. The ready to use fairness metrics that the tool offers are: selection parity, parity of exposer, and IGF (in group fairness). Selection parity measures whether the selection rates between the two groups in the top-k are equal. The parity of exposure measures whether a group's exposure is equal to the exposure of the other group at the top. Lower values mean the ranking offers equal representation/exposure among the groups. IGF is computed as the ratio between the lowest accepted score and the highest rejected score of a group at the top. Higher values mean more in group fairness.

Ranking Compare Annotate UI



Additionally, the comparison UI could be configured to collect annotations by asking the user to indicate which returned list of items is more suitable, given the query. In the option for displaying a ranking comparison, the UI will show in the top of the page the query, and below the two rankings to be compared.

The user can interact with the UI in the following ways:

- **Check Box:** depending on the task, the user can choose which ranking is more suitable for the displayed query.
- **Next Button:** the user can navigate to the next query using the Next Button.

For each user, the following interactions can be collected:

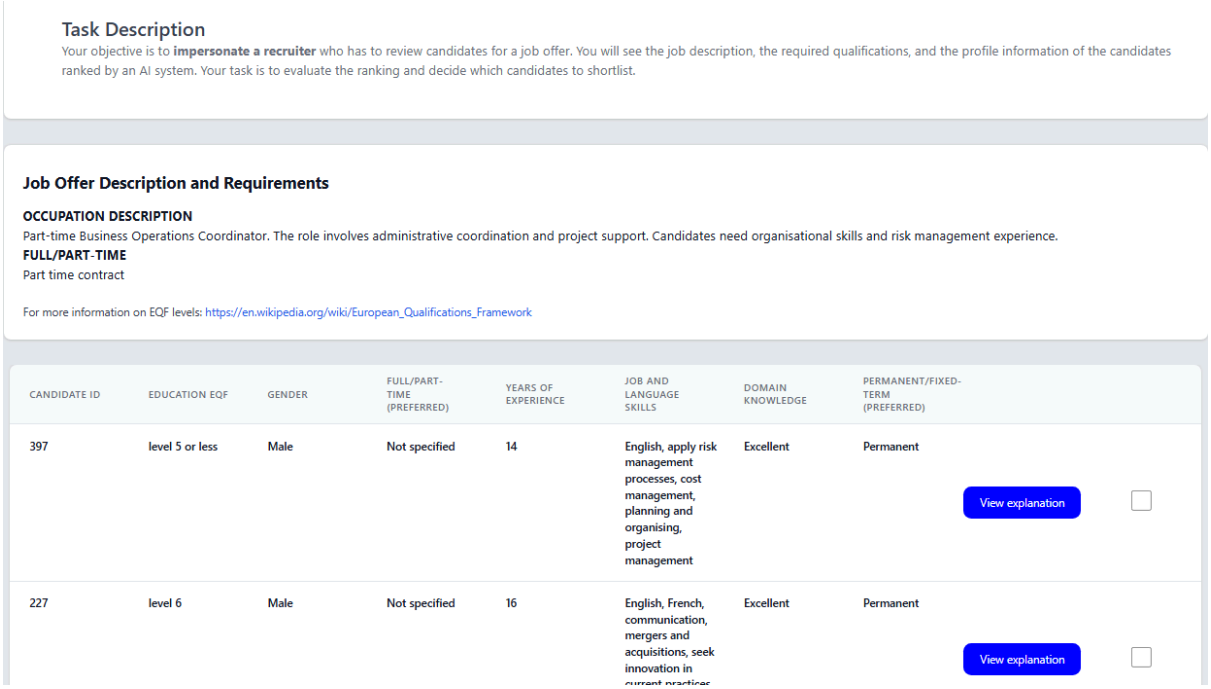
- **Best Ranking:** the chosen ranking by the user to be more suitable for the displayed query.

1.4 Explainability UI

XAnnorank builds upon the Display Interaction Annotate UI, supporting the evaluation of explanations for query-ranked-list and query-item-pair. Additionally, it supports displaying a built-in questionnaire component.

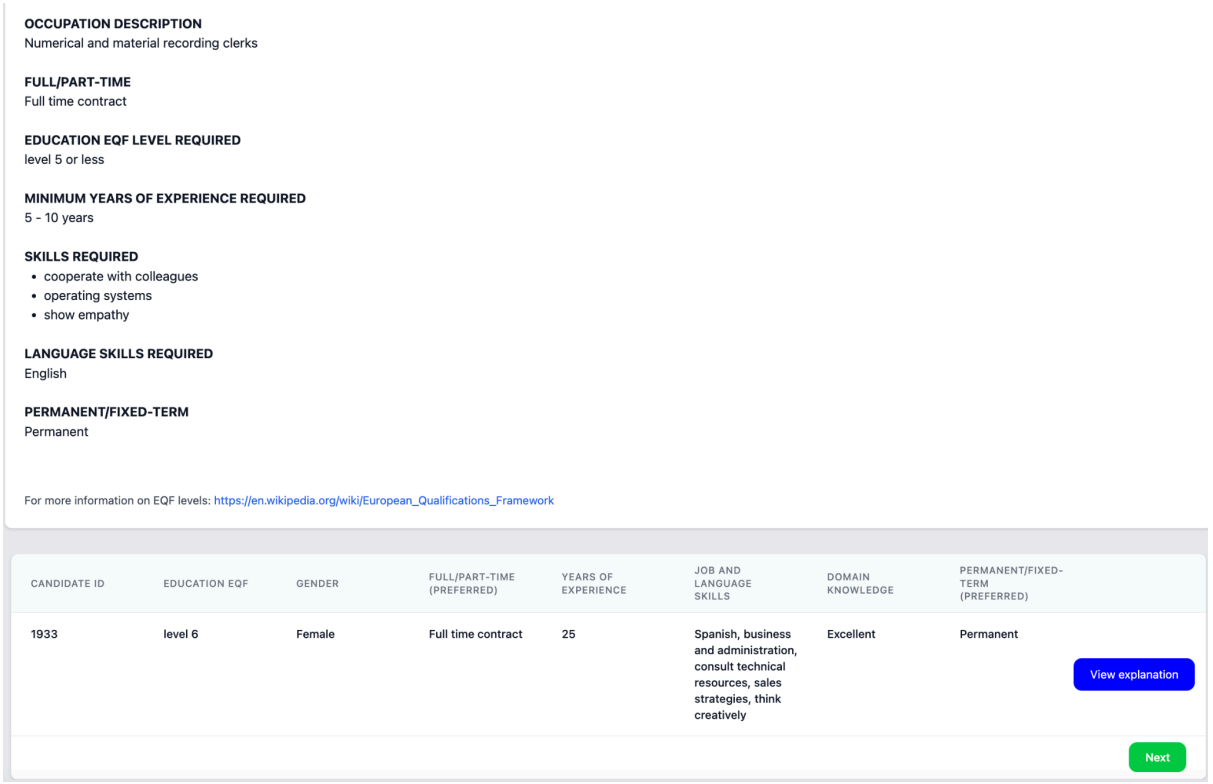
Query-Ranked-List

The figure below shows the Query-Ranked-List interface of XAnnorank in a recruitment setting, where recruiters are presented a ranked list of candidates who applied to the target job description. XAnnorank provides an additional explanation button - **View explanation** - that can show various explanation types.

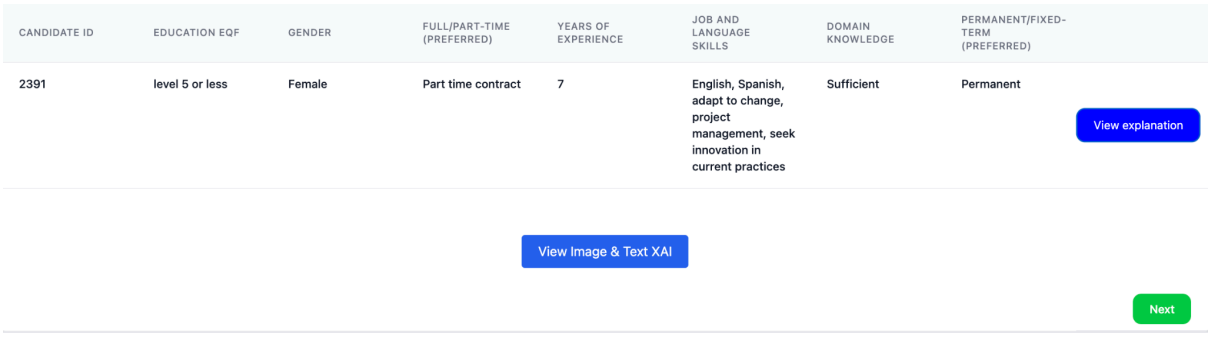


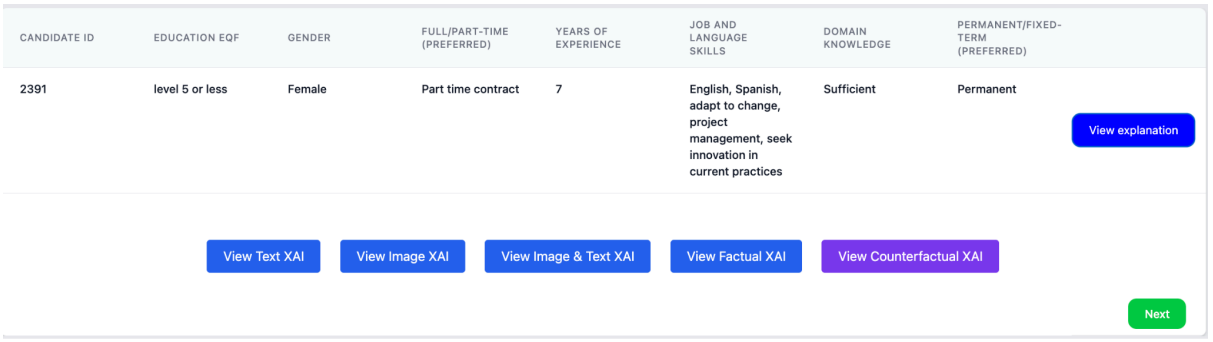
Query-Item-Pair

The Query-Item-Pair interface resembles the Query-Ranked-List interface in the figure above, but it displays only a single item, along with its corresponding metadata, and the corresponding query. In this recruitment demo, the Query-Item-Pair interface is presented to a candidate after applying to the corresponding job description.



View Explanation Button: When explanations are enabled, each item has a corresponding **View Explanation** button. By clicking this button, the user can access the explanation associated with the item. When clicked the following view will be available for the user to interact with:





One can configure which explanations are available in an experiment (e.g. all of available or just one specific type). The user can interact with the specific buttons for each explanation type. Multiple explanation views can be open at the same time.

For each user, the following interactions can be collected:

- **Clicks:** number of clicks for each document and XAI type on the View <xai_type> button, where <xai_type> can be “text”, “image”, “image_text”, “factual”, “counterfactual”
- **Timestamps:** for each click the timestamp of the click to open the expand and the timestamp of closing the expand are collected.
- **Top-k Documents:** the list of the selected documents in the order in which the user checked the box

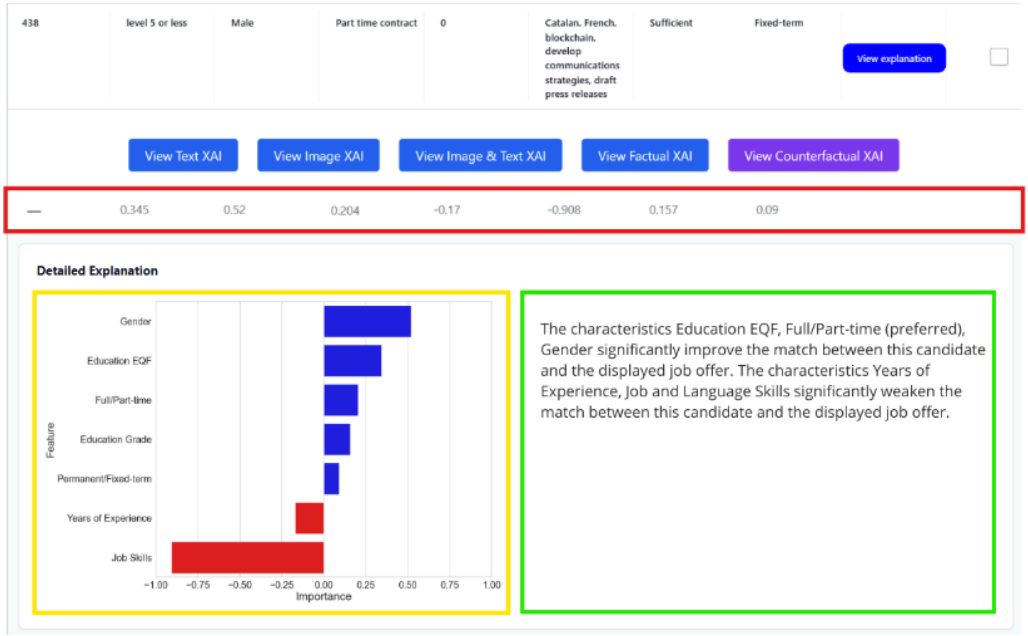
Explanations Types

XAnnorank supports the display of the Query-Ranked-List and the Query-Item-Pair interface with and without explanations. The explanation types supported include:

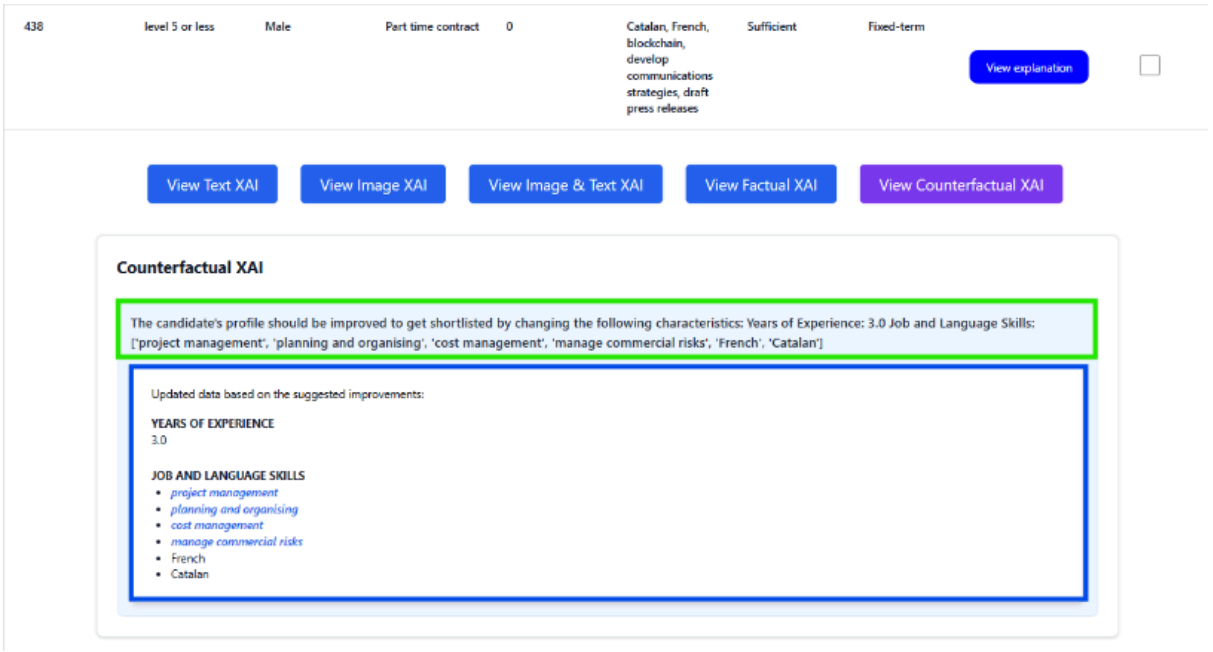
- **simple text**
- **image**
- **image-text combination**
- **factual explanations**
- **counterfactual explanations**

Textual explanations (green) present information in natural language, ranging from simple keyword-based descriptions to more advanced explanations. Visual explanations (yellow) display graphical elements such as charts. Factual explanations provide the feature-level importance to the ranking score based on which the candidates are ranked. They can be displayed either per feature (red) or using visual/text elements (yellow/green). Counterfactual explanations provide the minimal changes required for the candidate to improve its match to the job description. This can be displayed by showing the improved profile (blue) or using textual elements (green).

Factual

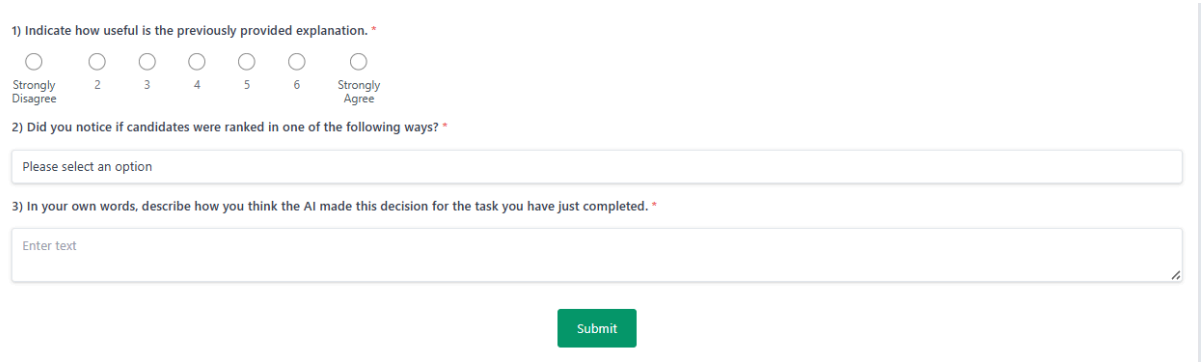


Counterfactual



Questionnaire

XAnnorank also supports the display of a built-in questionnaire component, which allows researchers to collect explicit feedback directly within the system. Multiple question types are supported, such as open-ended, multiple-choice and likert-scale. Questions can be marked as optional or mandatory. The figure below shows a few example questions from the questionnaire, illustrating the different supported question types. Additionally it supports the display of images.

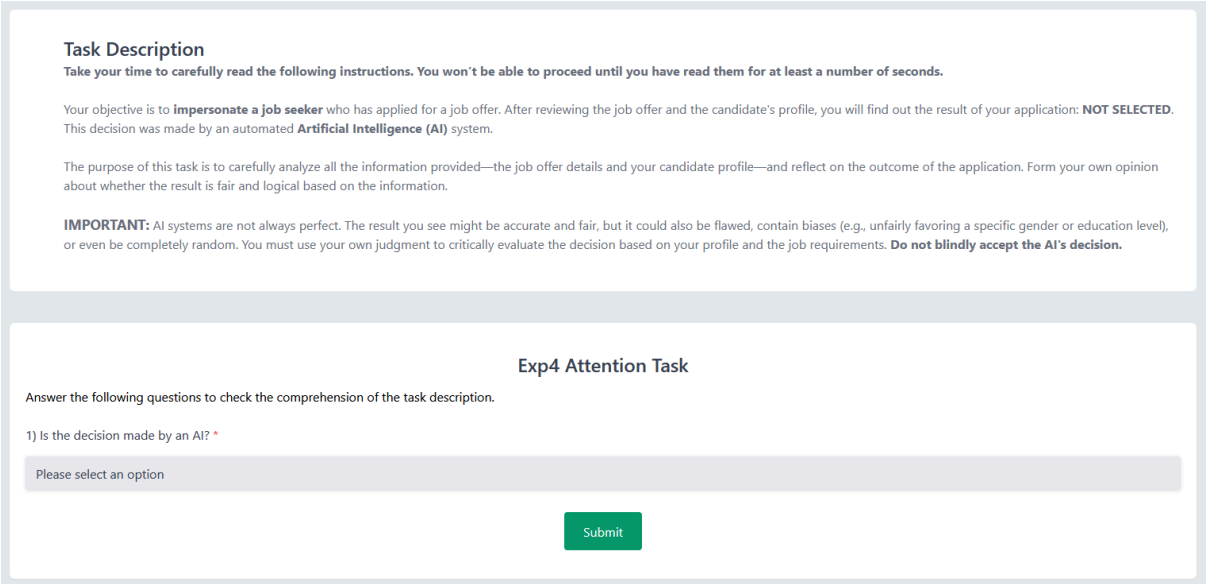


1.5 Attention Check

To ensure data integrity, attention checks can be added through the experimental pipeline. Attention checks supported are of two types:

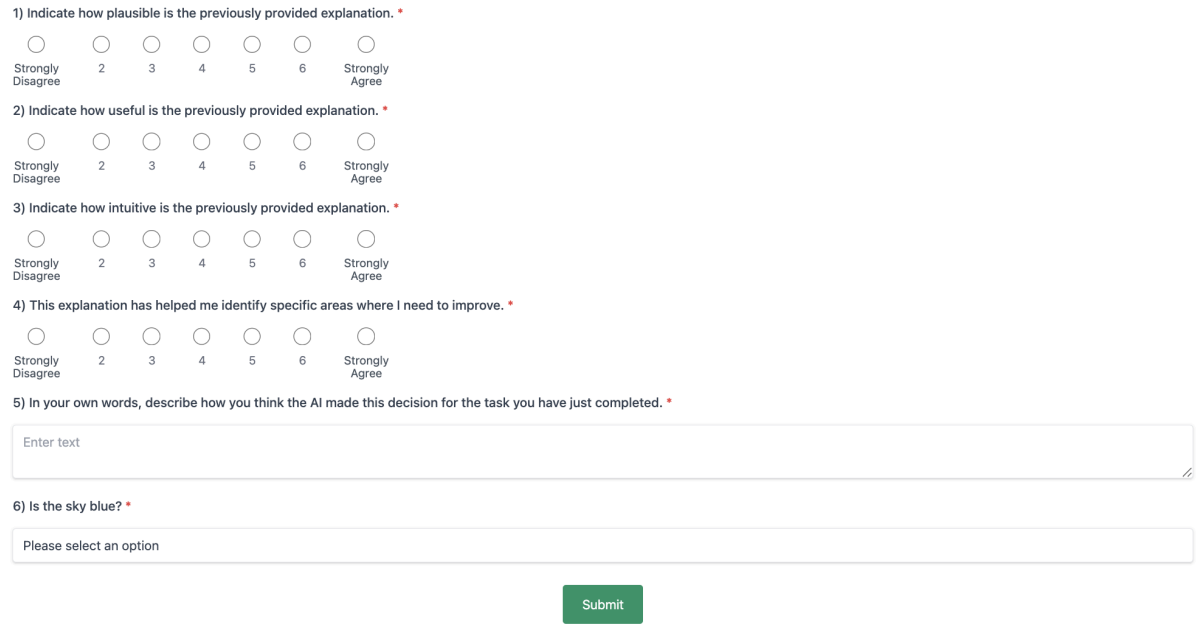
Comprehension Check

This type of attention check can be used to make sure that the user reads and understands the requirements of the task. The user can't go to the next task until they get the right answer. The tool will record the number of times a user imputed the wrong answer. Below is an example of the Comprehension Check.



Attention Questions

Attention questions can be inserted in the questionnaires to verify whether the user is paying attention while conducting the study. The user is allowed to continue the study even if answers are wrong. The number of failed attention questions is recorded in the database. This information can be later used by researchers to filter out users and improve the quality of the evaluation.



In this example, Question 6 is the **attention question.** It can be observed that it is listed within the questionnaire in between the actual questions used for the evaluation of the explanations.

2. Displaying a new Dataset

To use the UI with a new dataset, the following steps should be followed:

1. Under ./datasets, create a new folder called `<data_name>`/data and save the dataset there. Next, under ./datasets create a folder called experiments in which you can define the query-documents pairs to be displayed when running the UI. More details can be found in the following section: Experiment File.

2. Under ./src/data_readers create a python file `data_reader_<data_name>.py`. Inside create a class `DataReader<Data_name>`.

```
class DataReader<Data_name>(DataReader):

   def __init__(self, configs):
       super(DataReader<Data_name>,
self).__init__(configs=configs)

   def transform_data(self):
      # read the dataset file from self.data_path
            . . .

      # transform the dataset into two dataframes
      dataset_queries = pd.Dataframe(columns=['title', 'text'])
      dataset_documents = pd.Dataframe(columns=['docID',
      'query', 'score', 'group', ...])
            . . .

      # if needed split the dataframe for documents into
      data_train and data_test

      from sklearn.model_selection import train_test_split

      data_train, data_test =
      train_test_split(dataset_documents, test_size=0.2)
            . . .
      # if not needed return None for data_train
      data_test = dataset_documents
      data_train = None

      return dataset_queries, data_train, data_test
```

The `DataReader<Data_name>` should implement the `transform_data` method that should return the following:

- `dataframe_queries` - dataframe describing the queries. It should have the following mandatory columns:
  - `'title'` - this is the title of the query
  - `'text'` - this will be displayed in the UI

For example, if the query is a job description, the `'title'` will be the job title and the `'text'` will be the job description, which includes the job title. If the query is represented by a word/s set both `'title'` and `'text'` to the same value.

- `data_test` - dataframe describing the documents to be displayed.
- `data_train` - dataframe describing the documents to be used when training a ranker or a fairness intervention.

The formatted dataset will be saved under ./datasets/`<data_name>`/format_data

3. Under ./configs create json files following the naming conventions:
   - `config_shortlist_<data_name>.json` → to run the Interaction Annotate UI
   - `config_compare_<data_name>.json` → to run the Ranking Compare Visualise UI
   - `config_compare_annotate_<data_name>.json` → to run the Ranking Compare Annotate UI

   in which one should define the configuration to run the tool with. More details about the configuration file can be found in the following section: Configuration File.

Expected XAI Format

The dataframe used to display the XAI in the UI should follow a certain format. The column names of the dataframe should be:

- factual_xai
- counterfactual_xai
- image_xai
- text_xai

The xai data itself is expected in JSON format. It consists of a list of dictionaries, each containing fields that define the type of explanations to be shown, depending on the `ranking_type`. This supports the scenario of experimenting with various ranking functions. Be aware that each explanation type has a specific structure, which are presented below.

factual_xai

```
[
        {
        "data": {"Years of Experience": -0.34,
                 "Skills": 0.443},
        "factual_image": "plot.png",
        "factual_text": "Some textual explanation",
        "ranking_type": "rank1"
        },
        {...}
]
```

`"data"` - contains a dictionary of feature importance

`<key> : <feature importance>`

where `<key>` is the name of the feature and `<feature importance>` is the value of the feature importance determined by the factual xai method.

`"factual_image"` - path to the image accompanying the factual explanation
`"factual_text"` - description accompanying the factual explanation
`"ranking_type"` - name of the ranking file which correspond to a particular ranking function

counterfactual_xai

```
[
        {
        "data": {"Years of Experience": 5,
                  "Skills": ["*Python*",
                             "Dutch"]},
        "counterfactual_text": "Some textual explanation",
        "ranking_type": "rank1"
        },
        {...}
]
```

`"data"` - contains the updated item information/profile
`<key> : <value>`

where `<key>` is the name of the feature and `<value>` is the value of the feature. To highlight the skill(s) added in the improved profile for the counterfactual explanation, enclose those skill(s) in between asterisks (*****).

`"counterfactual_text"` - description accompanying the counterfactual explanation
`"ranking_type"` - name of the ranking file which correspond to a particular ranking function

text_xai

```
[
        {
        "text": "Some textual explanation",
        "ranking_type": "rank1"
        },
        {...}
]
```

`"text"` - simple text explanation
`"ranking_type"` - name of the ranking file which correspond to a particular ranking function

image_xai

```
[
        {
        "image": "plot.png",
        "ranking_type": "rank1"
        },
        {...}
]
```

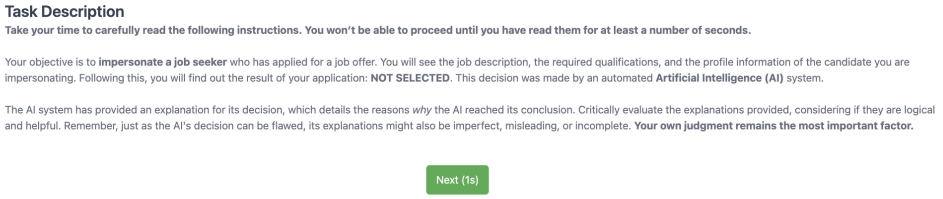
"image" – path to the image representing the explanation
"ranking_type" – name of the ranking file which correspond to a particular ranking function

3. Configuration File

Under ./configs create a josn file which represents the configurations with which to run the tool with the desired dataset.

"data_reader_class" – Configuration for the data.
    "name" – Specify the dataset name you want to run the tool with.
    "score" – Column to be used for ranking the documents.
    "group" – Column to be used by the fairness interventions representing the sensitive information of the documents (e.g. gender).
    "query" – Column representing the query to which the document is linked to. This should contain the same values found in dataframe_queries under 'title'.
    "docID" – Column representing the document IDs.

"ui_display_config" – Configuration for what to display in the UI.
    "highlmatight_ch" – Option to highlight the matching words of the query in the item's display field.
    "instructions_time" – enforced waiting time for the task description page before the button is enabled and the user can proceed to the next page



    "display_fields" – List of columns to display.
    "task_description" – guidelines of how the user should perform the assessment.
    "task_description_xai" – guidelines of how the user should perform the assessment with explanation.
    "exit_survey" – List of questions to be asked in the exit survey.
        "question" – The question to be displayed.
        "field" – The field name that will be used to store the value in the database.
        "options" – If a drop-down is wanted, list of options to be shown in the drop-down, otherwise set the value to "text" to display a free text input.
        "mandatory" – Boolean to be set if the field is mandatory to be completed by the user or not.

    Specific fields to declare for the Interaction Annotation UI:
    "shortlist_button" – Boolean value for whether to display the shortlist button.

"shortlist_select" - List representing the range [min, max] of top-k items the user has to select as being the most relevant for the given query.

"view_button" - Boolean value for whether to display or not the view (Explanation) button. For running XAnnoRank this is not required. This will be enabled/disabled depending on the "show_xai" defined in the experiment file for each task.

"view_fields" - List of columns to display when clicking on the view (Explanation) button. For running XAnnoRank this is not required.

Specific fields to declare for the Score Annotation UI:

"score_range" - range for the score bar (e.g [1,5]), with the first value being the start of the range and the second value being the last value from the range. The score bar will have 5 circles starting with values from 1 to 5.

Specific fields to declare for the Ranking Comparision UI:

"annotate" - Boolean to be set to true if wanting to run the Ranking Compare Annotate UI, or set to false if wanting to run Ranking Compare Visualise UI.

"avg_interaction" - Configuration for the interactions to be displayed.

"experiment_id" - Experiment ID out of which we want to visualise the collected interactions.

"interaction" - Interaction type to be displayed (e.g. "n_views" - clicks on the view button).

"display_metrics" - Configuration for metrics to be displayed.

"top_k" - Compute the metric at @k.

"utility_metrics" - List of utility metrics to be computed and displayed.

"fairness_metrics" - List of fairness metrics to be computed and displayed.

"attention_check" - Configuration to define which task is the attention check.

Definition of attention_check for AnnoRank:

"task" - The task to be considered as the attention check.

"correct_answer" - The correct answer expected for this task. It should be defined as list of documents for the Interaction Annotation UI. For the Score Annotation UI it should be the value of the correct label.

Definition of attention_check for XAnnoRank:

"limit" - Number of accepted failed attention checks. If exceeded the user is redirected to the end of the experiment, and get's returned a FAIL code.

"reload_tasks" - List containing the indices of the comprehension tasks.

"iaa" - Configuration for the inter-annotator agreement

"krippendorfs" - Boolean to be set to true to compute Krippendorf's IAA [6]

"cohens" – Boolean to be set to true to compute Cohen's Kappa IAA [5]

"weighted_cohens" – Boolean to be set to true to compute Weighted Cohen's Kappa [7]

"filter_per_task" – Boolean to be set to true if wanting to compute the IAA per task

"train_ranker_config" – Configuration for applying a ranker on the original ranking. If set to null, no ranker will be applied to the original ranking, otherwise, the documents will be ranked based on the predicted score by the defined ranker.

"name" – Name of the ranker to be applied.

"model_path" – Path to load a pre-trained model. If no pre-trained model is available, a new model will be saved at the specified path.

"settings" – List of configurations to run the ranker (specify your own fields used by the ranker class). The ranker class should be initialised with the settings specified.

"ranking_type" – Specify under which tag the ranking of the documents should be saved in the database.

- ranker_<name> - for running a ranking model

"pre/in/post_processing_config" – Configurations to apply a pre-processing fairness intervention on the data. If set to null, no ranker will be applied to the original ranking, otherwise, the documents will be ranked based on the predicted score by the defined ranker.

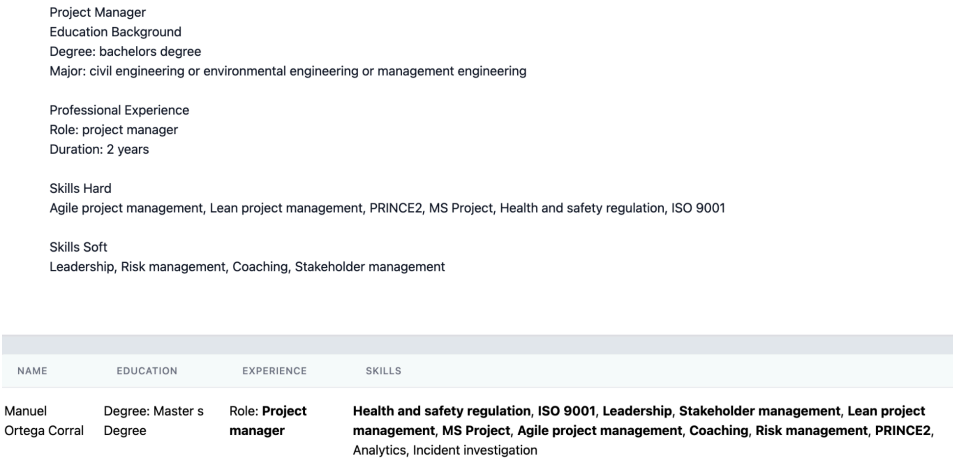
"name" – Name of the method to be applied.

"model_path" – Path to load a pre-trained model. If no pre-trained model is available, a new model will be saved at the specified path.

"settings" – List of configurations to run the method (specify your own fields used by the fairness intervention class). The fairness intervention class should be initialised with the settings specified.

"ranking_type" – Specify under which tag the ranking of the documents should be saved in the database. It is important to keep the following naming convention for the ranking type:

- preprocessing_<name> - for running a preprocessing fairness intervention.
- postprocessing_<name> - for running a postprocessing fairness intervention.
- inprocessing_<name> - for running an in-processing fairness intervention.



Example of the "highlight_match" functionality

4. Experiment File

Note: In XAnnoRank experiments are shown in the order defined in the file and not in random order. In the previous version of AnnoRank the experiments were displayed in random order.

Experiment File for the Interaction Annotation UI

"exp_id" - id of the experiment.
"description" - description of the experiment to run.
"tasks" - list of assessments.
    "query_title" - the title of the query to be displayed.
    "setting" - an extra requirement that the user should take into account when doing the assessment.
    "ranking_type" - the order in which to display the ranking of the documents. It should match the "ranking_type" defined in the configuration file.
        "ranking_type" could have the following values:
            "original" - sorted by the "score".
            "ranker_<name>" - sorted by the predicted score of a pre-trained model.
            "pre/in/postprocessing_<name>" - displays the ranking generated by the fairness intervention.

Experiment File for the Score Annotation UI

"exp_id" - id of the experiment.
"description" - description of the experiment to run.
"tasks" - list of assessments.
    "query_title" - the title of the query to be displayed.
    "setting" - an extra requirement that the user should take into account when doing the assessment.
    "index" - the index of the item to be annotated.

Experiment File for the Ranking Comparison UI

"exp_id" - id of the experiment.
"description" - description of the experiment to run.
"tasks" - list of assessments.
- "query_title" - the title of the query to be displayed.
- "ranking_type_1" - the order in which to display the ranking of the documents that is displayed on the left side. It should match the "ranking_type" defined in the configuration file.
- "ranking_type_2" - the order in which to display the ranking of the documents that is displayed on the right side. It should match the "ranking_type" defined in the configuration file.
  - "ranking_type" could have the following values:
    - "original" - sorted by the "score".
    - "ranker_<name>" - sorted by the predicted score of a pre-trained model.
    - "pre/in/postprocessing_<name>" - displays the ranking generated by the fairness intervention.

Experiment File for XAnnoRank

"exp_id" - id of the experiment. Order as defined in file.
Experiments can be of type: XAI/ not XAI query-ranking, XAI/ not XAI query-item, questionnaire, attention check, and comprehension check.
"description" - description of the experiment to run.
"tasks" - list of assessments.
- "query_title" - the title of the query to be displayed.
- "ranking_type" - the order in which to display the ranking of the documents. It should match the "ranking_type" defined in the configuration file.
  - "ranking_type" could have the following values:
    - "rank<n>" – predefined ranking order
    - "form" – when task is form
- "cand_idx" – if the experiment is: XAI/ not XAI query-item, it is the index of the item to be displayed. If this is set only that item will be displayed, otherwise the full list defined in the ranking file corresponding to "ranking_type".
- "description" – description of the task.
- "setting" - an extra requirement that the user should take into account when doing the assessment.
- "show_xai" - type of XAI to display for this task – false if no XAI should be displayed. It can have the following values:
  - "false" – no XAI will be displayed
  - "Factual" – shows the factual per feature importance + image + text – if available
  - "Counterfactual" – shows the counterfactual text + updated item data
  - "Text" shows just text
  - "Image" shows just an image

"Image_text" shows both an image and text
"True" – It will show all available XAI

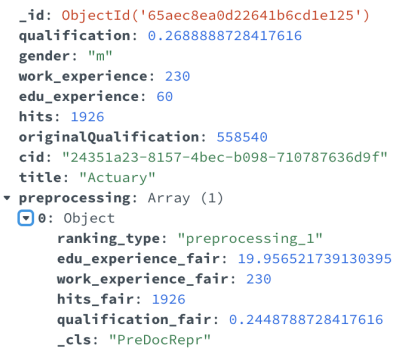
"questionnaire" - array of items, where each item is a question. Each item has the following values:

"question" – The question to be displayed. This can be text or an image
"label" – Optional. If an image is displayed you can add text to go with the image by using this field for the text.
"field" – The field name that will be used to store the value in the database.
"options" – If a drop-down is wanted, list of options to be shown in the drop-down, otherwise set the value to "text" to display a free text input.
"mandatory" – Boolean to be set if the field is mandatory to be completed by the user or not.
"correct_answer" - If the task, is attention check, it contains the correct answer.
If query_title is set to be an actual query from the database, it means that the tool will first show the task corresponding to the query, followed by the questionnaire. Otherwise, it will display the questionnaire as a task.

5. Database

The tool works with a MongoDB database. The diagram below describes the collections present in the database. For each dataset, a new database is created automatically together with the required collections. The name of the created dataset will be the name specified in the configuration file as the "data_reader_class". Below the use of each collection is described:

- documents - stores the dataframe describing the data to be displayed, meaning the data stored in data_test. On top of the fields displayed in the diagram, the tool automatically adds the rest of the columns present in data_test. The preprocessing field stores a list with preprocessed values given the configurations of the ranking_type.



- queries - stores the dataframe describing the queries, meaning the data stored in dataframe_queries.

```
_id: ObjectId('65aec8ea0d22641b6cd1e124')
title: "Actuary"
text: "Actuary"
```

```
_id: ObjectId('65b038abf3c7907e51baf8cf')
title: "project_manager"
text: "Project Manager
       Education Background
       Degree: bachelors degree
       Major: civil engineering or environmental engineering or management

       Professional Experience
       Role: project manager
       Duration: 2 years
```


- `dataset` - stores the query-rankings pairs. For each pair, it stores the id of the query and a list of rankings. Each ranking stores an ordered list of the IDs of the documents linked to this query. The lists are ordered based on the `ranking_type` that was defined in the configuration file.


```
_id: ObjectId('65aec8ec0d22641b6cd1e308')
query: "65aec8ea0d22641b6cd1e109"
▾ rankings: Array (6)
  ▾ 0: Object
    ▸ docs: Array (8)
      ranking_type: "original"
  ▾ 1: Object
    ▸ docs: Array (8)
      ranking_type: "preprocessing_1"
  ▾ 2: Object
    ▸ docs: Array (8)
      ranking_type: "ranker_1:qualification__qualification"
  ▾ 3: Object
    ▸ docs: Array (8)
      ranking_type: "ranker_1:qualification_fair__qualification_fair"
  ▾ 4: Object
    ▸ docs: Array (8)
      ranking_type: "postprocessing_1"
  ▾ 5: Object
    ▸ docs: Array (8)
      ranking_type: "postprocessing_2"
```


- `experiments` - stores the data defined in the experiment file, including the list of tasks to be displayed when running the tool. On the left there is an example belonging to an experiment file used to run the Interaction Annotation UI, while on the right an example belonging to an experiment file used to run the Ranking Compare UI. The corresponding tasks can be seen under the next point: on the left for the Interaction Annotation UI and on the right for Ranking Compare UI.


```
_id: ObjectId('65aec8ed0d22641b6cd1e349')
_exp_id: "4"
_description: "test experiment"
▸ tasks: Array (4)
    0: "65aec8ed0d22641b6cd1e345"
    1: "65aec8ed0d22641b6cd1e346"
```

```
_id: ObjectId('65aec8ed0d22641b6cd1e344')
_exp_id: "3"
_description: "compare rankings side by side"
▾ tasks: Array (3)
    0: "65aec8ed0d22641b6cd1e341"
    1: "65aec8ed0d22641b6cd1e342"
```


- `tasks/tasks_compare/tasks_score` - stores the data described in the experiment file for each task, together with the ID of the query-ranking pair stored in the dataset collection. On the left, there is an example belonging to collection `tasks`, while in the middle an example belonging to collection `tasks_compare`, and on the right an example belonging to collection `tasks_score`.


```
_id: ObjectId('65aec8ed0d22641b6cd1e345')
data: "65aec8ec0d22641b6cd1e30b"
ranking_type: "postprocessing_1"
query_title: "Actuary"
```

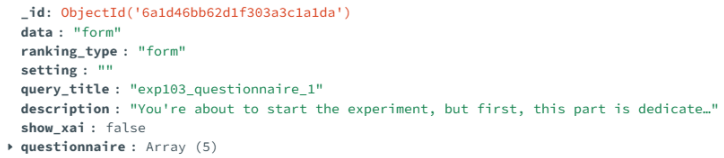
```
_id: ObjectId('65aec8ed0d22641b6cd1e346')
data: "65aec8ec0d22641b6cd1e30b"
ranking_type: "original"
query_title: "Actuary"
```

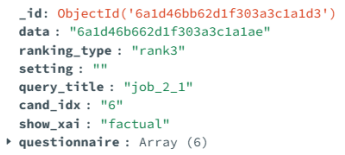
```
_id: ObjectId('65aec8ed0d22641b6cd1e341')
data: "65aec8ec0d22641b6cd1e30b"
ranking_type_1: "original"
ranking_type_2: "postprocessing_1"
query_title: "Actuary"
```

```
_id: ObjectId('65aec8ed0d22641b6cd1e342')
data: "65aec8ec0d22641b6cd1e30b"
ranking_type_1: "original"
ranking_type_2: "ranker_1:qualification__qualification"
query_title: "Actuary"
```

```
_id: ObjectId('65c0e3fa193007bae2a8c54b')
data: "65c0df6ad4bfa872c5efdff9"
index: "1"
setting: ""
query_title: "Actuary"
ranking_type: "original"
```

For XAnnorank the different tasks are stored in `tasks`. The left image is an example of the task questionnaire, the right image is an example of XAI query-ranking. Different types of XAI can be displayed. The field show_xai contains the type of explanation that will be displayed.





- `user` - stores the ID provided by the user on the first page of the tool and the list of tasks the user interacted with in the tool. As the user interacts with the tool, for each task viewed we store the ID of the tasks together with the list of collected interactions for each document.

`exit_form` - stores the answers to the exit form.

`tasks_visited` - stores the list of tasks already visited by the user, together with the collected interactions. It is useful for the resuming of the experiment.

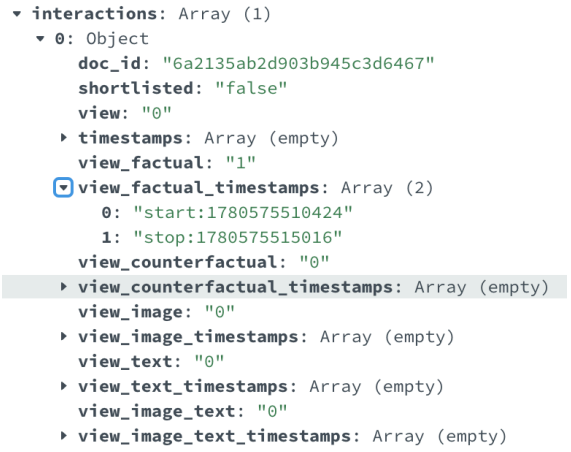
`interactions_compare` - stores the interactions for the Ranking Compare UI

`interactions_score` - stores the interactions for the Ranking Score UI

`interactions` - stores the interactions for the Interaction Ranking Annotate UI, including the interactions with the XAI part

`order_checkbox` - the order in which the shortlisted top-k items are clicked

`form_submission` - stores the answers to the questionnaires



`view_<xai_type>` – collects the number of clicks for the View <xai_type> Button of XAnnoRank

`view_<xai_type>_timestamps` – collects the timestamps of the View <xai_type> Button of XAnnoRank

`view` – collects the number of clicks for the View Button of AnnoRank

`timestamps` – collects the timestamps of the View Button of AnnoRank

`shortlisted` – true/false depending on whether the document was shortlisted or not

Example of questionnaires results:

Example of exit form results:

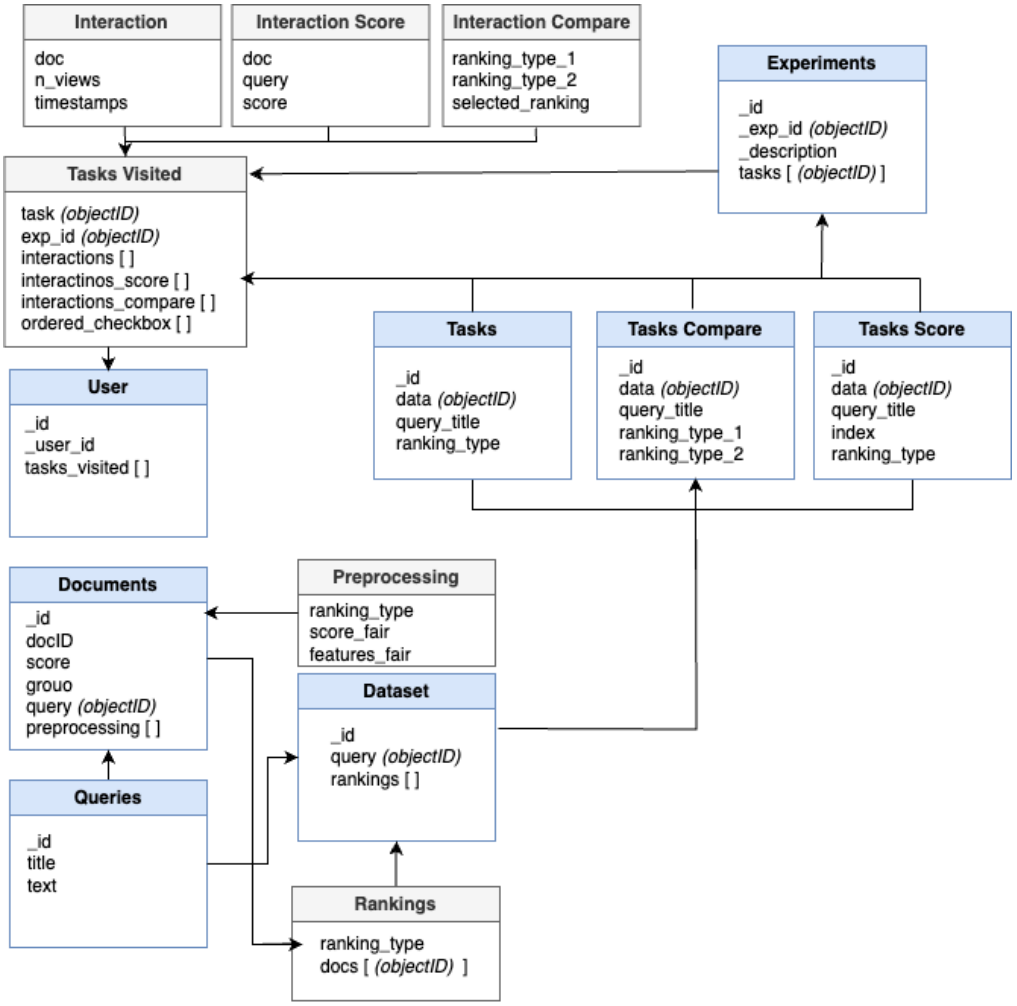
Storing of attention check

`_attention_check` – number of failed attention checks

`_reload_attention_check` – number of failed comprehension tasks

Export Data

The collected data can be exported by either running the following command: *mongoexport --host="<mongo_hostname>:<mongo_port>" --collection=<collection_name> --db=<db_name> --out=<output>.json* or from the MongoDB Compass interface[1] as either json or csv files.



[1] https://www.mongodb.com/produdcts/tools/compass

Inter-annotator agreement

Anno Rank offers the possibility of computing inter-annotator agreement on the exported annotations. So far, Anno Rank supports computing three IAA's modes (Krippendorf's Alpha [7], Cohen's Kappa [5], Weighted Cohen's Kappa [6]), using the Python *agreement* library[2]. This feature can be activated through a configuration file, for example:

```
"iaa": {
    "krippendorfs": true,
    "cohens": true,
    "weighted_cohens": true,
    "filter_per_task": true }
```

, where one has to set the desired IAA metric configuration (`"krippendorfs", "cohens", "weighted_cohens"`) to 'true' for computing the metric or 'false' for not computing the metric.

Enabling `"filter_per_task"` configuration will group the annotations based on their *experiment_id*, *task_id*, and *doc_id*. If set to false, it will compute the user agreements for every annotating task in the database for a particular dataset.

To compute the IAA metrics for the "shortlisting" feedback collected with the Interaction Annotation UI, the IAA metrics configuration must be included in the c*onfig_shortlist_<dataset_name>.json*. To compute the IAA metrics for the "label" feedback collected with the Score Annotation UI, the IAA metrics configuration should be included in the *config_annotate_score_<dataset_name>.json*.

To generate these metrics, the user can execute this command at any time: `docker exec annorank-container conda run -n tool_ui python /app/src/evaluate/iaa_metrics.py --dataset <dataset_name>`, as long as the MongoDB container is still up. The <*dataset_name*> should be the same as to the `config["data_reader_class"]["name"]` in the same configuration file. After executing the docker exec command above, the output of the IAA calculation will be stored in the */output/iaa_metrics.jsonl*.

Example when `"filter_per_task"` is True:

```
{
    "annotate": {
        "filters": {
            "task_id": "0",
            "exp_id": "2",
            "doc_id": "6655e6b4e5a0a19c08e948c0"
        },
        "iaa_metrics": {
            "krippendorffs": 0.6,
            "cohens": 0.6,
            "weighted_cohens": 0.6
        },
        "error_message": "No error"
    }
```

[2] https://pypi.org/project/agreement/

```
        },
    {
            "ranking": {
                "filters": {
                    "task_id": "0",
                    "exp_id": "1",
                    "doc_id": "6655e6b4e5a0a19c08e948bf"
                },
                "iaa_metrics": {
                    "krippendorffs": 0.5,
                    "cohens": 0.5,
                    "weighted_cohens": 0.5
                },
                "error_message": "No error"
            }
        }
```

Example when `"filter_per_task"` is False:

```
{
      "ranking": {
          "filters": {
              "no_filter": true
          },
          "iaa_metrics": {
              "krippendorffs": 1.0,
              "cohens": 1.0,
              "weighted_cohens": 1.0
          },
          "error_message": "No error"
      }
  }
```

The value ranges from 0 to 1, where 0 is perfect disagreement about a specific pair query and document and its annotation score, where 1 is ideal agreement for more than one annotator.

Attention Check - AnnoRank

Anno Rank offers the possibility to define an attention check task. In the database the user will have a flag indicating whether the user passed the attention check or not. Using the attention check one could discard the users that did not pass it.

Example of attention check defined for the Flickr dataset:

Interaction Annotate UI:

```
"attention_check": {
   "task": {
```

```
      "query_title": "1000344755.jpg",
      "ranking_type": "original"
    },
    "correct_answer": [16, 17, 18]
  }
```

Score Annotate UI:

```
"attention_check": {
 "task": {
   "query_title": "1000344755.jpg",
   "index": "2",
   "ranking_type": "original"
 },
 "correct_answer": "Relevant"
}
```

The "`correct_answer`" indicates the list of documents that the user should shortlist in order to pass the attention check. If the collected feedback from the user differs from the one indicated under "`correct_answer`" the user will have the attention check failed in the database.

6. Additional Support

6.1 Ranklib Rankers

Ready to Use

The tool offers support for using any model implemented using the ranklib library[3]. It is to be considered that the ranklib library requires java to be installed in the docker container, thus, on the first run it might take longer for the app to start as it first needs to install java. It can be used by defining in the config file the following:

```
"train_ranker_config": {
 "name": "Ranklib",
 "model_path": "./dataset/<data_name>/models/Ranklib", // path to
save/load the model

 "settings": [{
     "features": ["feature_1", "feature_2", "feature_3"], // list
of features to be considered during training
     "pos_th": 0, // threshold to consider relevant documents
     "rel_max": 500, // maximum relevance that is assigned based
on the 'score' column
     "ranker": "RankNet", // ranker name
     "ranker_id": 1, // ranker id

     "metric": "NDCG", // metric to evaluate
     "top_k": 10, // evaluate at top-k
     "lr": 0.000001, // learning rate
```

```
"epochs": 20, //number of epochs

"train_data":["original"] // define the train data,
"test_data": ["original"] //define the test data,
"ranking_type": "ranker_1" }]}
```

For more information on how to use ranklib rankers check the ranklib documentation: https://sourceforge.net/p/lemur/wiki/RankLib%20How%20to%20use//.

The `"train_data"` and `"test_data"` can be set to the following values:

- `"original"` - the ground truth is set to be the original score
- `<ranking_type>` - if the ground truth is set to be the score computed using a fairness intervention

The `train` method saves the input files in the format expected by the ranklib library under ./dataset/`<data_name>`/models/Ranklib/`<train_data>_<test_data>`. The model and predictions are saved under ./dataset/`<data_name>`/models/Ranklib/`<train_data>_<test_data>`/ranklib_experiments/`<ranker_name>`.

The output of the `predict` method returns a new dataframe that is in the same format as the original one with an appended column representing the predicted score. The column with the predicted relevance should follow the convention `<train_column>__<test_column>`. For example if the train and test data is set to be "original", the predicted score column should be `"score"__"score"`, where `"score"` is the value defined in the config file under the `"score"` field. If the train and test data are pre-processed by a fairness intervention the predicted score column should be `"score"_fair__"score"_fair`.

The new ranking based on the predicted score is saved in the MonogDB database in the collection `dataset`, in the field rankings. Given the config presented above, the ranking type of saved in the database will be set to `ranker_1:"score"__"score"`. If a preprocessing fairness intervention is applied on the train/test data, the ranking type saved in the database will be set to `ranker_1:preprocessing_1:"score"_fair__"score_fair"`

Add a new ranker

In order to add a new ranker the following steps should be followed:

1. Under ./src/rankers create the following python file: ranker_`<ranker_name>`.py
   Inside the python file create the class that implementes the ranker method:

```
class <ranker_name>Ranker(Ranker):
    def __init__(self, configs, data_configs,
    model_path):
```

```
super().__init__(configs,data_configs,
model_path)
```

`model_path` - path to save/load the model
`configs` - dictionary of hyperparameters needed to run the ranker. This is defined in the config file as “settings”.
`data_configs` - dictionary of configs defined for the `data_reader_class`. This is needed to be able to access the required columns by the fairness intervention.

2. Implement the methods defined in the parent class `Ranker`, which can be found in ./src/fairness_interventions/ranker.py.

```
def train_model(self, data_train, data_test, experiment)
```
`data_train` - data to train the model
`data_test` - data to evaluate the model during training
`experiment` - tuple containing the <train_ranking_type> and <test_ranking_type> defined in the configuration file.

The `train` method should save the trained model at the following path:
`self.model_path/<train_ranking_type>__<test_ranking_type>`.

```
def predict(self, data, experiment)
```
`data` - data to run the model on and generate the predicted ranking
`experiment` - tuple containing the <train_ranking_type> and <test_ranking_type> defined in the configuration file.

The `predict` method should load the model from `self.model_path/<train_ranking_type>__<test_ranking_type>` and apply it on the `data`.

The method should return a new dataframe that is in the same format as data with an appended column representing the predicted score. The column with the predicted score should follow the convention `<train_score_column>__<test_score_column>`.

For example if the `<train_ranking_type>` and `<test_ranking_type>` is set to be “original”, the predicted score column should be `”score”__”score”`. If the train and test data are pre-processed by a fairness intervention the predicted score column should be `”score”_fair__”score”_fair`, where `“score”` is the value defined in the config file under the `“score”` field.

3. If needed any files related to the fairness method can be saved under ./src/rankers/modules/<ranker_name>

4. Define the new ranker in ./src/constants in the dictionary containing the rankers.

5. Using the new ranker:

```
"train_ranker_config": {
      "name": "<ranker_name>" // same as the key defined in
the dictionary,
      "model_path":
"./dataset/<data_name>/models/<ranker_name>", // path to
save/load the model
      "settings": [{
            // define any configs needed to run the ranker
            "features": ["feature_1", "feature_2",
"feature_3"], // list of features to be considered during
training

            "train_data":["original"], // define the train
data
            "test_data": ["original"], //define the test data
            "ranking_type": "ranker_1"
            }]}
```

6.2 Fairness Interventions

Ready to Use

The tool supports applying:

- post-processing fairness intervention: FA*IR[2]. It can be used by defining in the config file the following:

```
"post_processing_config": {
 "name": "FA*IR",
 "model_path": "", //empty as there is no model to save
 "settings": [{
      "k": 10,
      "p": 0.7,
      "alpha": 0.1
      "ranking_type": "postprocessing_1"}]
 }
```

- pre-processing fairness intervention: CIF-Rank[1]. It can be used by defining in the config file the following:

```
"pre_processing_config": {
      "name": "CIFRank",
      "model_path": "./dataset/<data_name>/models/CIFRank", // path to
save/load the model

      "settings": [{
```

```
            "pos_th": 0, //threshold to consider positive documents
            "control": "group_value", //control group, the group
           towards which we convert all the data in a counterfactual
           world
            "features": ["field_1", "field_2", "field_3"]//list of
      features (columns from the dataframe) to be considered as
      mediators
}]
}
```

It is to be considered that the fairness interventions might require additional libraries to be installed in the docker container, thus, on the first run it might take longer for the app to start as it first needs to install the required libraries.

Add a new fairness method

In order to add a new fairness intervention the following steps should be followed:

1. Under ./src/fairnes_interventions create the following python file: fairness_method_<method_name>.py

   Inside the python file create the class that implementes the fairness method:

   ```
   class <method_name>(FairnessMethod):
         def __init__(self, configs, data_configs, model_path):
               super().__init__(configs, data_configs, model_path)
   ```

   model_path - path to save/load the model
   configs - dictionary of hyperparameters needed to run the fairness method. This is defined in the config file as "settings".
   data_configs - dictionary of configs defined for the data_reader_class. This is needed to be able to access the required columns by the fairness intervention.

2. Implement the methods defined in the parent class FairnessMethod, which can be found in ./src/fairness_interventions/fairness_method.py.

   ```
   def train_model(self, data_train)
      data_train - data to train the model
   ```

   The train_model method should save the fairness intervention method to self.model_path.

   ```
   def generate_fair_data(self, data):
      data - data to be used to generate the fair data
   ```

   The generate_fair_data method should return a new dataframe that has the same columns as data, to which the fair columns are added following the name convention <column_name>_fair.

   For example, when running a pre-processing fairness intervention like CIF-Rank[1], both the features defined under "features" and the score define

under "score" will be changed. The fair columns should be "score"_fair, where "score" is the value defined in the config file under the "score" field, and same for all the features defined under "features". When running a post-processing intervention like FA*IR[2], which re-ranks the candidates, then the fair column should represent the new ranking, thus, the name of the returned column is rank_fair.

3. If needed any files related to the fairness method can be saved under ./src/fairness_interventions/modules/<method_name>

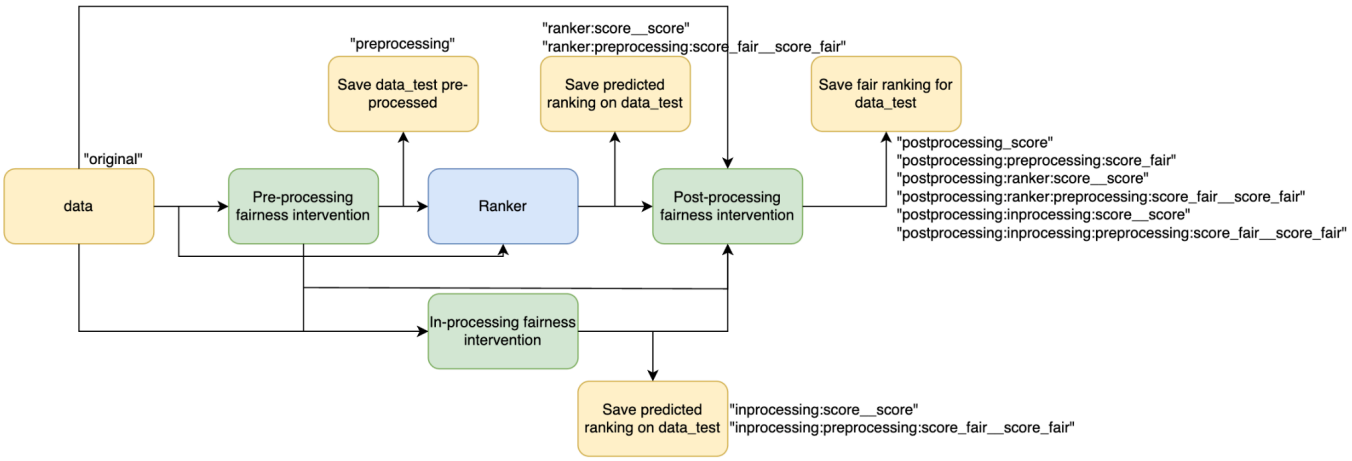
4. Define the new fairness method in ./src/constants in the dictionary containing fairness methods.

5. Using the new fairness method:

```
      "name": "<method_name>" // same as the key defined in the dictionary,
      "model_path": "./dataset/<data_name>/models/<method_name>", // path to
save/load the model
      "settings": [{
            // define any configs needed to run the fairness method
            "train_data":["original"], // define the train data for an
in-processing method
            "test_data": ["original"], //define the test data for an
in-processing method
            "ranking_type": // define the ranking type as according to the
naming convention specified in the Configuration File section
            }]
```

The figure below describes how the fairness interventions can be applied on the data and how their outputs can be used to train a ranking model, if the configuration file has enabled the use of both the fairness interventions and the use of a ranking model. The **Pre-processing fairness intervention** is applied on the data and saved in the database. As mentioned before, the fair values are saved in the documents collection, while the fair ranking is saved in the dataset collection. The **Ranker**, the **In-processing fairness intervention** and the **Post-processing fairness intervention** can be trained/tested on either the pre-processed data or on the original data. This can be set in the configuration file using the field "train_data" and "test_data". The predicted rankings are saved in the database in the collection dataset. The post-processing method can be applied on any kind of ranking, including the ranking based on the original "score" column, the ranking based on the output of the fairness interventions or of the ranker. The predicted rankings are

saved in the database under the `dataset` collection.



Given the following example of a configuration for the post-processing intervention using FA*IR[2]:

```
"post_processing_config": {
 "name": "FA*IR",
 "model_path": "",
 "settings": [{
      "k": 10,
      "p": 0.7,
      "alpha": 0.1,

      "test_data": ["original", "preprocessing_1",
      "ranker_1:preprocessing_1:qualification_fair__qualification_fair",
      "ranker_1:qualification__qualification"],

      "ranking_type": "postprocessing_1"}]
```

The post-processing fairness intervention will be applied on the following rankings:

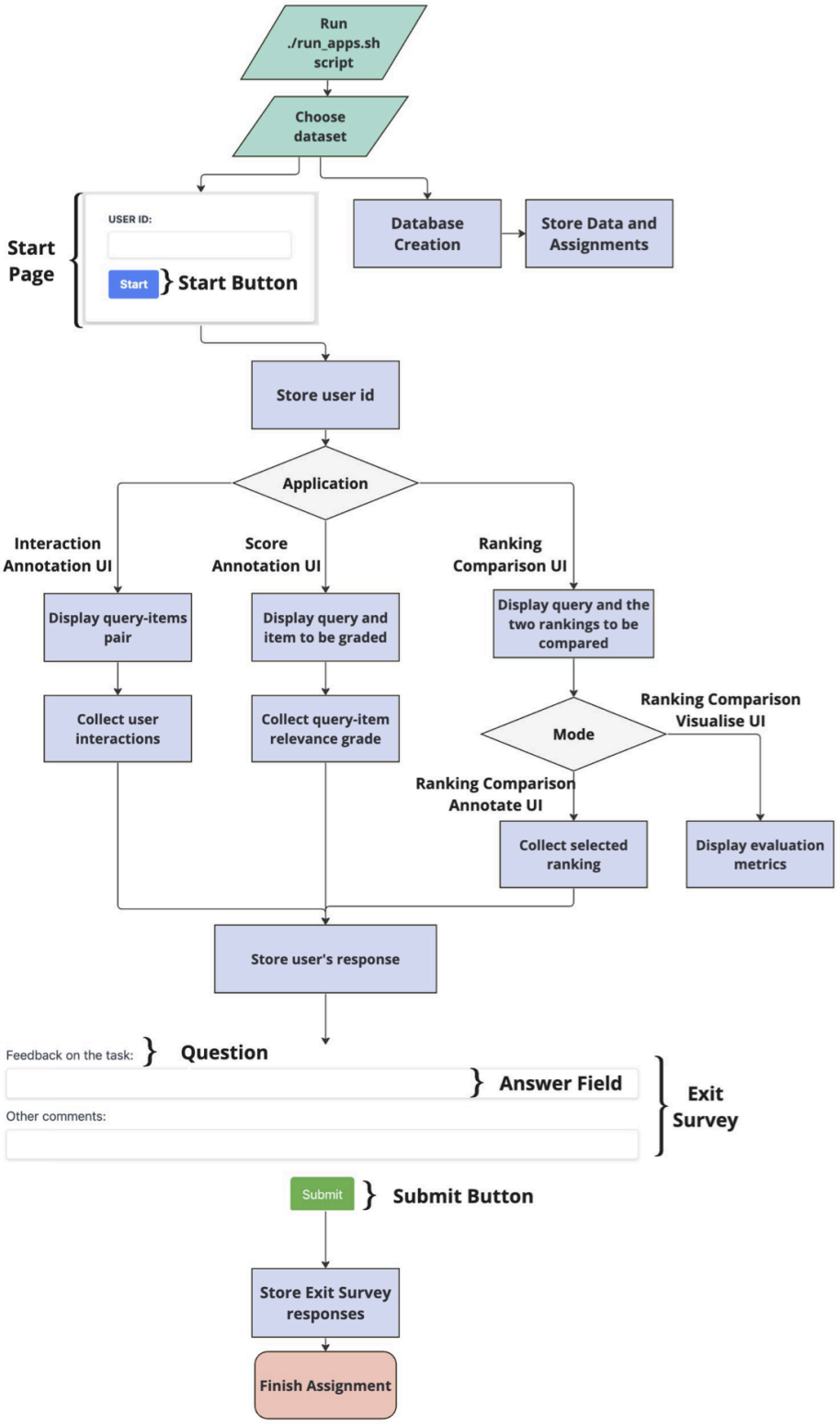
- `"original"` - the ranking produced by column “score”
- `"preprocessing_1"` - the ranking produced by the pre-processed score column (`qualification_fair`)
- `"ranker_1:preprocessing_1:qualification_fair__qualification_fair"` - the ranking produced by `ranker_1` which was trained on the `qualification_fair` column produced by the pre-processing method defined as `preprocessing_1`.
- `"ranker_1:qualification__qualification"` - the ranking produced by `ranker_1` which was trained on the `qualification` column.

The ranking_type of the ranking saved in the database are defined as it follows for the above use cases:

- `postprocessing_1:qualification`
- `postprocessing_1:qualification_fair`
- `postprocessing_1:ranker_1:preprocessing_1:qualification_fair__qualification_fair`
- `postprocessing_1:ranker_1:qualification__qualification`

7. Run the tool

AnnoRank is designed to be easy to use. We provide easy to run examples, that can be used to tailor the app to specific needs, as well as a step by step tutorial. The figure below depicts the workflow of the AnnoRank application.



Requirements:

AnnoRank utilizes docker containers to handle our applications and database. The *docker-compose* command directs the machine to construct the necessary containers and their dependencies as outlined in the *docker-compose* file. This includes installing the necessary python packages to run the app as well as other requirements such as Java to be able to make use of the Ranklib library, or R to be able to run the pre-processing fairness intervention. Depending on the use of the app one can comment out or add the dependecises in the *docker-compose* file. To make use of other python packages the env.yml file needs to be updated. Thus, the AnnoRank web apps can be effortlessly deployed on any web hosting server.

- Install Docker by following the steps presented here: https://www.digitalocean.com/community/tutorials/how-to-install-and-use-docker-on-ubuntu-18-04 Select the operating system of the device you want to run the app on and proceed with the steps indicated.

- Install Docker Desktop: https://www.docker.com/products/docker-desktop/ to be able to monitor the docker instances.

- Install MongoDB Compass: https://www.mongodb.com/products/tools/compass to view the dataset created and its collections. The connection should be set as mongodb://<IP>:27017. <IP> should be set to IP address for of the machine where the docker container is running.

- If you are using Windows make sure you have WSL2. Allow WSL2 usage in docker settings.

Starting the app is easy and straightforward as one needs to only define the desired configuration files and run the script *run_apps.sh*. Add the paths to the configuration files needed to run the app in: **apps.docker.sh.** If the script cannot be executed, this may be caused by Windows line endings. This issue can be resolved by converting the files to Unix format using: **dos2unix run_apps.sh** and **dos2unix apps_docker.sh**.

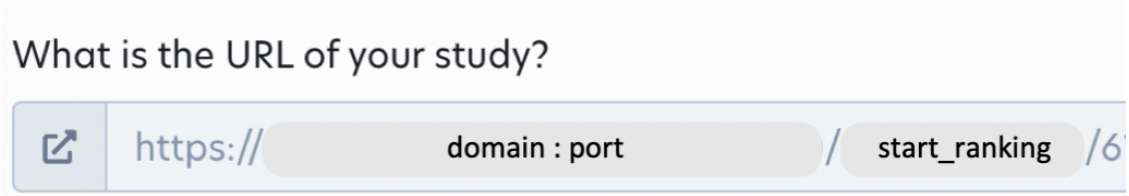
The script will lunch the creation of the docker container, the population of the database with the specified dataset, and the web apps. Please note that whenever the dataset is modified, the **format_data** folder within the dataset directory and existing Docker containers should be removed before rerunning the script. After starting the app, the web apps will be accessible at the following URLs:

- **Interaction Annotation UI:** http://localhost:5000/start_ranking/<exp_id>
- **Score Annotation UI:** http://localhost:5003/start_annotate/<exp_id>
- **Ranking Comparison Visualise UI:** http://localhost:5001/start_compare/<exp_id>
- **Ranking Comparison Annotate UI:** http://localhost:5002/start_compare_annotate/<exp_id>

The ports where the apps are running are defined in the *docker-compose* file, if needed one can change the ports to other values. The <*exp_id*> is the ID of the assignment to be displayed to the users which was defined in the experiment file and stored in the collection experiment.

Crowd-sourcing platforms

AnnoRank can be easily used with crowd-sourcing platforms. For example to use AnnoRank with the Prolific platform one needs to only create a study inside the prolific platform and set the URL of the study to the URL of the UI tool:



The example above makes use of the Interaction Annotation UI showing to the user the assignment corresponding to ID 6. The user will be registered in the database given the Prolific ID provided by the platform.

8. Ready to run Use Cases

AnnoRank offers tutorials for using pre-built datasets, fairness interventions, and ranker models. However, it can be tailored to meet various information retrieval needs.
We describe the three use cases below.

8.1 Recruitment (XAnnorank)

Below we show XAnnorank in a recruitment setting. Explainability (XAI) is especially important in high-stakes domains such as recruitment, where recommendations may have significant ethical and legal implications. However, XAnnorank can also be used in other settings like banking, healthcare, insurance and criminal justice, where explainability is equally important. XAnnoRank can be adapted to a wide variety of experiments by supporting different datasets, ranking models, and researcher-specified settings.

This example supports two user perspectives:

1. That of a **candidate**, where the user impersonates a job seeker who was not selected by the AI system and is asked to critically evaluate the AI's decision based on their profile and the job requirements.
2. That of a **recruiter**, where the user acts as a recruiter reviewing the AI-ranked candidate list and must shortlist a defined number of candidates.

In both cases, XAnnoRank can present the ranked list with and without AI-generated explanations, collecting both implicit interaction data and explicit feedback through questionnaires.

Run the Recruitment Use Case

Run the following script and type "findhr":

```
cd Annorank
./run_apps.sh
```

To access the tool for the Demo:

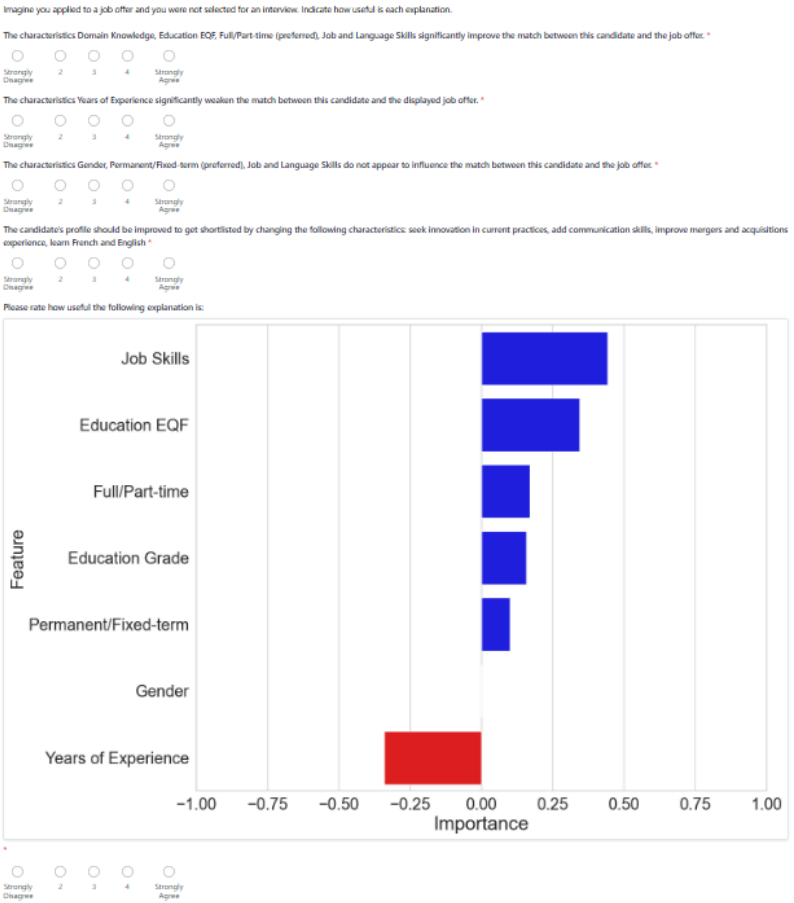
- Candidate side: http://localhost:5005/start_ranking_XAI/<exp_id>
  - Exp_id: 101, 102, 103, 104
- Recruiter side: http://localhost:5004/start_ranking_XAI/<exp_id>
  - Exp_id: 1, 2, 3, 4

The experiments are defined as follows:

Candidate Side

http://localhost:5005/start_ranking_XAI/101

This experiment is composed only of questionnaires. In our recruitment demonstration we show to a candidate various explanations and ask the candidate to evaluate their usefulness. XAnnoRank supports both the display of text, images and images with text in the questionnaires.



http://localhost:5005/start_ranking_XAI/102

In this experiment we ask the user to impersonate a job seeker. The user is then presented with the various job descriptions and their profile. The user is asked to interact with the UI presenting in each task a different type of explanation. In the follow-up questionnaire the user is asked to evaluate the previously seen explanation type. In this way XAnnoRank can be used to compare various types of explanation methods.

http://localhost:5005/start_ranking_XAI/103
In this experiment we ask the user to impersonate a job seeker. The user is then presented with the various job descriptions and their profile. The user is asked to interact with the UI which this time does not show any explanation. In the follow-up questionnaire the user is asked to evaluate whether the reason for rejection is clear. With such a set-up XAnnoRank can be used to conduct A-B test experiments, where one pool of users is presented with the explanations, and the other without explanations.

http://localhost:5005/start_ranking_XAI/104
In this experiment the user is first presented with the job description and their profile without explanations, in the second part of the study they are presented with the explanation. Each task is followed by a questionnaire evaluating the previously seen explanations and interaction with the UI. In this way one can evaluate the impact of various explanation methods on the user's perception and behaviour.

Recruiter Side - experiments are similar to the ones defined above, with the difference that they should be shown to a recruiter, and the recruiter is presented with the job description together with the list of candidates who applied to that particular job offer.

http://localhost:5004/start_ranking_XAI/1
This experiment is composed only of questionnaires. The recruiter is presented with various explanations and asked to evaluate them.

http://localhost:5004/start_ranking_XAI/2
This experiment shows various types of explanations followed by a questionnaire. The recruiter is asked to choose the best candidates to be shortlisted.

http://localhost:5004/start_ranking_XAI/3
The recruiter is asked to choose the best candidates, but without the extra information provided by the explanations. This gives the opportunity to run an A-B test study by showing to a pool of candidates the task without XAI, and to another pool the task with XAI. This is useful in understanding how explanations impact the recruitment process and the recruiter’s behaviour.

http://localhost:5004/start_ranking_XAI/4
In this experiment the recruiter is first presented with the recruitment task without the XAI, and in the second part with the XAI.

Configuration files are available at the following path: ./AnnoRank/configs/findhr_tutorial

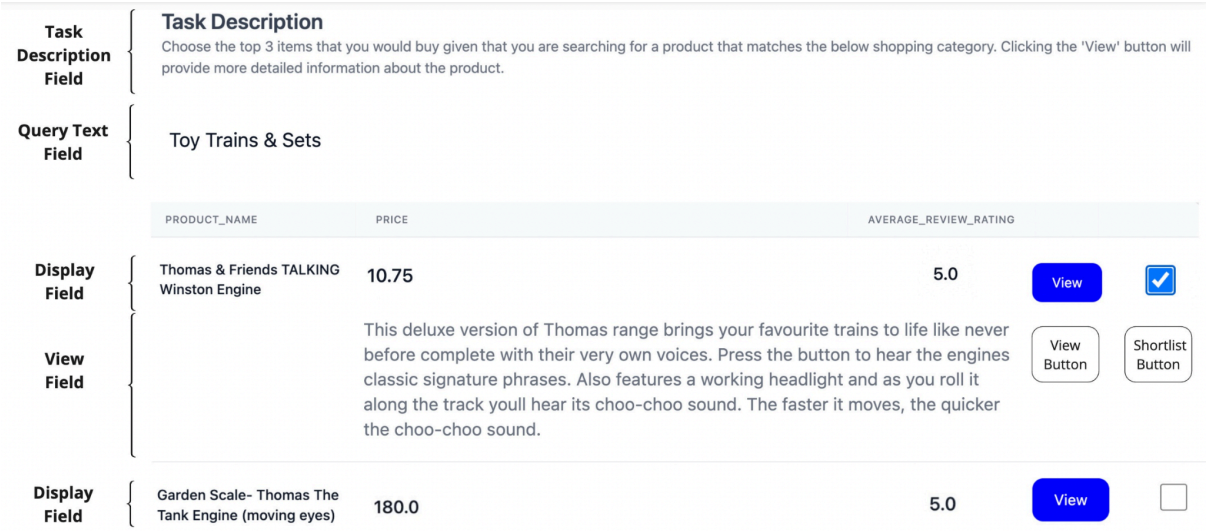
The dataset is available at the following path: ./AnnoRank/dataset/findhr

The example provided consists of made-up data and does not reflect the data of real people. Candidate profiles are described by education level (EQF), gender, working time preference, years of experience, job and language skills, and domain knowledge. Gender serves as the protected attribute for fairness analysis. Each candidate file includes four types of XAI output.

The experiment files are available the following path: ./AnnoRank/dataset/findhr/experiments

Each task specifies a query_title, a ranking_type determining which ranking model is used, and a show_xai field controlling which explanation type is presented: true for all explanations, or a specific type such as "factual", "counterfactual", "text", "image", or "image_text". A per-task questionnaire can be defined to collect explicit feedback immediately after each ranking task

8.2 Retail (Annorank)



AnnoRank can be valuable in enhancing existing datasets with user interactions that are essential for the training of a recommender system. As a possible concrete use-case for the Interaction Annotation UI, one could consider the collection of user interactions for a retail dataset. Understanding user preferences is essential to improve user satisfaction in the retail industry. The Amazon e-commerce dataset[3] serves as a prominent example of this application. The configuration file depicted in Listing 1 illustrates the straightforward integration of the Amazon dataset features into the AnnoRank UI. For example, the

3 https://www.kaggle.com/code/residentmario/exploring-toy-products-on-amazon/input?select=amazon_co-ecommerce_sample.csv

average_review_rating is valuable for ranking items in the dataset as it correlates well with user satisfaction.

Additionally, the *amazon_category_and_sub_category* feature is a strong candidate for query due to its association with multiple data points in the dataset. For example, a query for *toys* would retrieve all items related to *toys*. The *manufacturer* feature can be used as a *group* configuration option if fairness to manufacturers is desired, allowing the tool to retrieve items in a rank that is fair to the manufacturing group. For example, not all items from Manufacturer X will always be at the top of the rank; it will also include other manufacturing groups in the top list. By setting *product_name*, *price*, and *average_review_rating* as the *display_field* configuration option, we observe in that these features will be displayed in the display_field, while the *product_information* will be presented in the *view_field* that the annotator user will see after they click the *View* button.

Listing 1: Example of configuration file for running the Interaction Annotation UI with the Amazon dataset.

```
{
  "data_reader_class": {
    "name": "amazon",
    "score": "average_review_rating",
    "group": "manufacturer",
    "query": "amazon_category_and_sub_category",
    "docID": "product_name"
  },

   "ui_display_config": {
    "view_button": true,
    "shortlist_button": true,
    "shortlist_select": [3,3],
    "display_fields": ["product_name", "product_description",
...],
    "view_fields": ["product_information"],
    "task_description": "Select top 3 items ...",
  }
}
```

Run the Amazon Use Case

Run the following script and type "amazon":

```
cd UI-Tool-main
./run_apps.sh
```

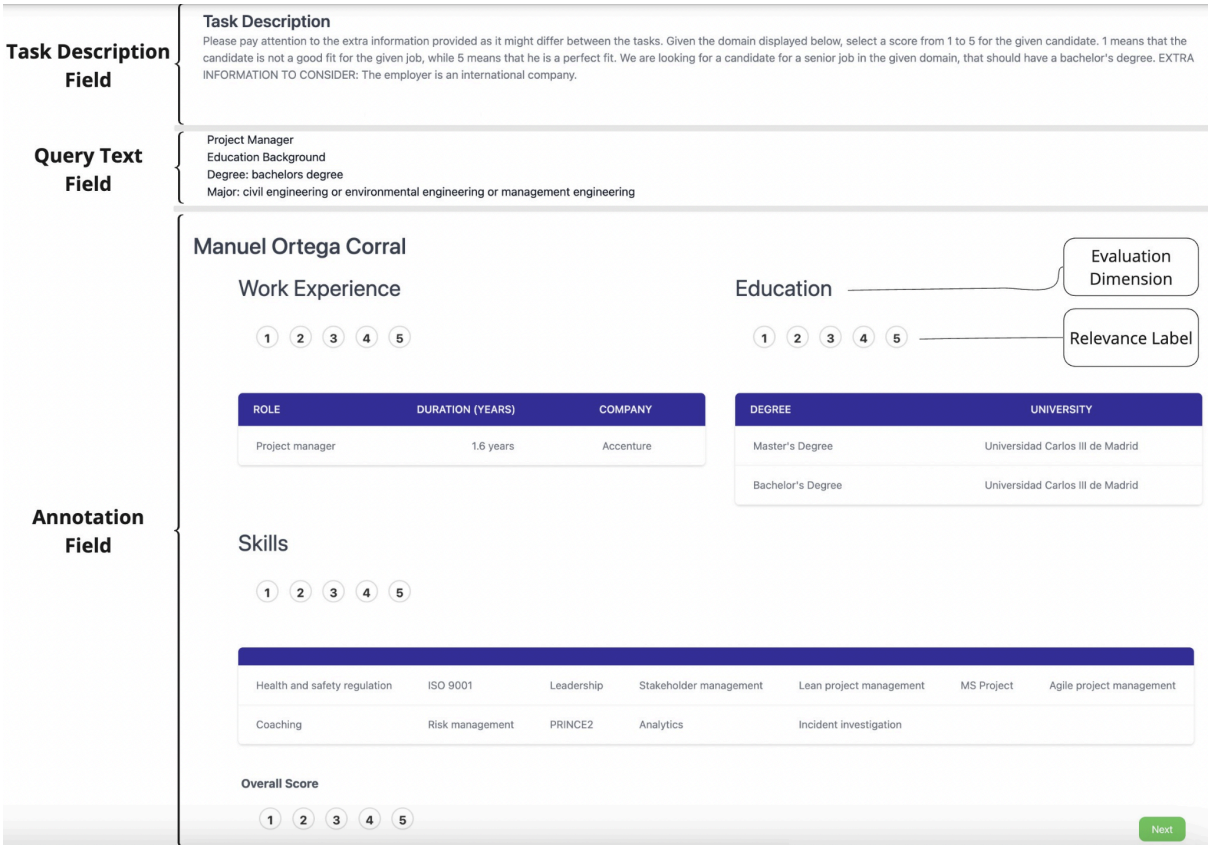
Before accessing the links you need to wait for the app to finish the install and start. The Amazon dataset is now available at the following links:

- To access the Interaction Annotation UI go to the following link: http://localhost:5000/start_ranking/3

- To access the Score Annotate UI go to the following link: http://localhost:5003/start_annotate/1
- To access the Ranking Comparison Visualise UI go to the following link: http://localhost:5001/start_compare/2
- To access the Ranking Comparison Annotate UI go to the following link: http://localhost:5002/start_compare_annotate/2

8.3 Recruitment System (Annorank)

Graded relevance judgments can be applied to both a single dimension or to more dimensions which can be defined according to the features of the dataset. For example, a candidate applying for a job can be evaluated based on three dimensions: the skills, the education, and the work experience required by the job description. Thus, the Score Annotation UI can be customized to a use-case where the user must annotate every defined dimension of the item.



The figure above shows an example of how the Score Annotation UI can be customized for a recruitment scenario, where the user is asked to annotate not only the candidate's overall fit to the given job requirements, but also each individual dimension.

The example provided consists of made-up data and does not reflect the data of a real person. In the example provided, when the annotation assessment starts, the user is tasked with evaluating the candidate with respect to the query ``Customer Service Representative''

and the corresponding job requirements. The user must grade the candidate's skills, education, and work experience given the job requirements. In the example, the ``Customer Service Representative'' position requires a candidate with at least one year of relevant experience. Additionally, the user must take into account the additional requirement defined in the Task Description field. As mentioned above, such additional requirements can be defined in the experiment configuration file as shown in Listing 2. The user must indicate the relevance of the candidate's experience by selecting a score ranging from 1 (very irrelevant) to 5 (very relevant). Once the user submits their assessment of relevance for all defined dimensions, the feedback is stored in the database and the experiment continues with the next assessment.

Listing 2: Example of configuration experiment file for running the Score Annotation UI with a recruitment dataset.

```
{
  "exp_id": 1,
  "description": "description of the experiment",
  "tasks": [
    {
      "query_title": "customer service representative",
      "setting": "Consider the employer to be an"
      "international company.",
      "ranking_type": "original"
    }, ... ]
}
```

Run the Recruitment Use Case

Run the following script and type "cvs":

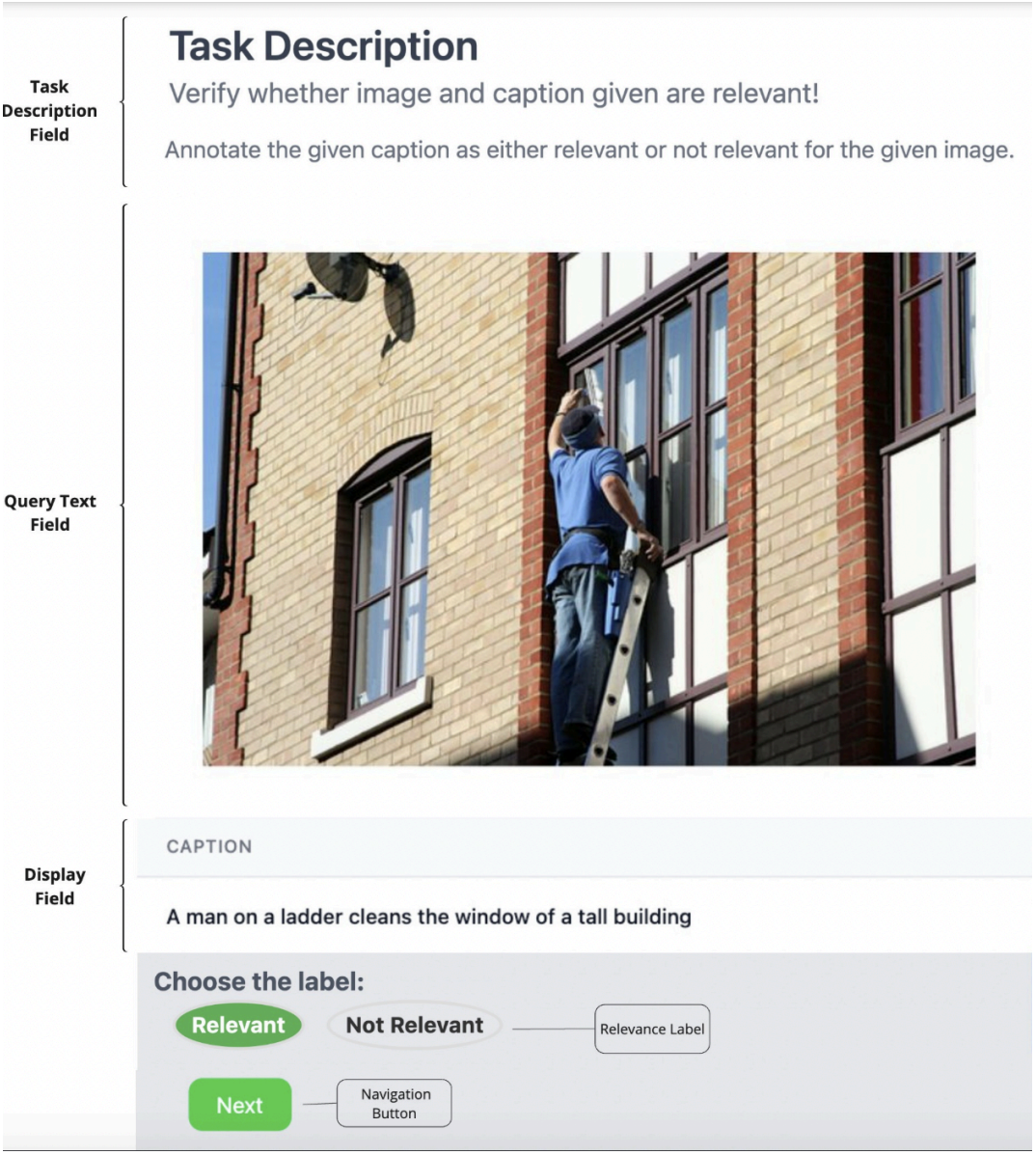
```
cd UI-Tool-main
./run_apps.sh
```

Before accessing the links you need to wait for the app to finish the install and start. In order to use the Score Annotate UI with multi dimension annotation adapted for the recruitment use-case, change in "/templates/index_annotate_documents" the line "{% include 'doc_annotate_profile_template.html' %}" with the following: {% include 'doc_annotate_profile_template_recruitment.html' %}.

The Recruitment dataset is now available at the following links:

- To access the Interaction Annotation UI go to the following link: http://localhost:5000/start_ranking/1
- To access the Ranking Comparison Visualise UI go to the following link: http://localhost:5001/start_compare/3
- To access the Ranking Comparison Annotate UI go to the following link: http://localhost:5002/start_compare_annotate/3
- To access the Score Annotate UI go to the following link: http://localhost:5003/start_annotate/2

8.4 Multimodal (Annorank)



Although information retrieval (IR) systems were initially developed for text retrieval, they have since advanced to handle a wide range of multimedia data, which includes text, audio, images, and video. With this in mind, the tool can be used for annotating the relevance of both images and captions. To demonstrate this functionality, we used a sample dataset from Flickr30k[4], which is used to address the problem of visual notation similarity within linguistic expressions captured in the captions provided for individual images \cite{young-etal-2014-image}. The figure above illustrates an example of how the Score Annotation UI can be used to annotate the relevance of a caption to an image query. The

[4] https://www.kaggle.com/datasets/hsankesara/flickr-image-dataset

image is stored in base64 format in our implementation for easier embedding to the HTML \emph{<img>} tag.

Run the Multimodal Use Case

Run the following script and type "flickr":

```
cd UI-Tool-main
./run_apps.sh
```

Before accessing the links you need to wait for the app to finish the install and start.

The Flickr dataset is now available at the following links:

- To access the Interaction Annotation UI go to the following link: http://localhost:5000/start_ranking/2
- To access the Ranking Comparison Visualise UI go to the following link: http://localhost:5001/start_compare/3
- To access the Ranking Comparison Annotate UI go to the following link: http://localhost:5002/start_compare_annotate/3
- To access the Score Annotate UI go to the following link: http://localhost:5003/start_annotate/1

9. Tutorial

Example on the XING dataset.

1. Download the XING dataset from: https://github.com/MilkaLichtblau/xing_dataset
2. Create the following folder ./datasets/xing/data and save the dataset there
3. The data reader implemented for this dataset can be found in the following python file ./src/data_readers/data_reader_xing.py.

4. The experiment files can be found at the following locations:
   ./datasets/xing/experiments/experiment_shortlist.json → to run the Interaction Annotation UI
   ./datasets/xing/experiments/experiment_compare.json → to run the Ranking Comparison UI
   ./datasets/xing/experiments/experiment_annotate_score.json → to run the Score Annotate UI

5. The config files can be found under ./configs/xing_tutorial/

- `config_create_db_xing.json` → configuration used to add data in the database the data

  The configuration files specifies that a pre-processing fairness intervention and a post-processing fairness intervention is applied on the data. It also specifies to train a

ranker on the data and store its prediction in the database. An example of how the data is stored in the database can be seen below.

On the left it can be seen that for each document and for each fairness intervention/ranker the predictions are saved in the database. On the right it can be seen that the ordering of all defined configuration is stored in the `dataset` collection.

- `config_shortlist_xing.json` → to run the Interaction Annotate UI

  This configuration file should create the following UI:

  In the configuration file, the `"shortlist_button"` is enabled together with the `"view_button"`. The `"shortlist_select"` field defines what is the minimum and the maximum items that the user needs to select as being the top most relevant items given the query. Under the "display_fields" you can define which fields to be shown first to the user, while under the `"view_fields"` you can define which fields to be shown when clicking on the "View" button. The `"exit_survey"` field defines what questions the exit survey should contain at the end of the assessments.

- `config_compare_annotate_xing.json` → to run the Ranking Compare Annotate UI

  Similarly, one can define which fields to be displayed and what the exit survey should contain. It is important to set `"annotate"` to be true, otherwise, no user ID will be collected and no checkboxes to select the most suitable ranking given the query and the requirements will not be present.

- `config_compare_xing.json` → to run the Ranking Compare Visualise UI

  Similarly, one can define which fields to be displayed. It is important to set `"annotate"` to false. In the configuration file one can select which metrics to be displayed for each ranking and which interaction data.

- `config_annotate_score_xing.json` → to run the Annotate Score UI

  Similarly, one can define which fields to be displayed. It is important to set the range for the scores to be displayed using the field `"score_range"`.

6. Requirements:
   Install Docker by following the steps presented here: https://www.digitalocean.com/community/tutorials/how-to-install-and-use-docker-on-ubuntu-18-04 Select the operating system of the device you want to run the app on and proceed with the steps indicated.

Install Docker Desktop: https://www.docker.com/products/docker-desktop/

Install MongoDB Compass: https://www.mongodb.com/products/tools/compass to view the dataset created and its collections. The connection should be set as mongodb://<IP>:27017. <IP> should be set to IP address for of the machine where the docker container is running.

7. Run the following script **run_apps.sh**

8. To access the Interaction Annotation UI go to the following link: http://localhost:5000/start_ranking/4

   To access the Ranking Comparison Visualise UI go to the following link: http://localhost:5001/start_compare/3

   To access the Ranking Comparison Annotate UI go to the following link: http://localhost:5002/start_compare_annotate/3

   To access the Score Annotate UI go to the following link: http://localhost:5003/start_annotate/5

Repository Structure

→ **configs**
- contains examples of configuration files for the supported datasets

→ **dataset**
- contains the supported datasets from the examples provided in the documentation

→ **external_resources**
  → **Anno_Rank_Documentation.pdf**
  - the documentation of the tool

  → **Usability_Study_Anno_Rank.pdf**
  - the usability study conducted for the tool

→ **flask_app**
  → **app_ranking_XAI.py**
  - runs the Interaction Annotate UI with XAnnoRank

  → **app_annotate_document.py**
  - runs the Score Annotate UI

  → **app_ranking.py**
  - runs the Interaction Annotate UI

  → **app_ranking_compare.py**
  - runs the Ranking Comparision UI (researchers only)

  → **app_ranking_compare_annotate.py**
  - runs the Ranking Comparison UI for annotators

  → **database.py**
  - contains the structure of the database

  → **db_create_pipeline.py**

- contains the script which is adding the formatted data in the database. It runs as well the ranking models and fairness interventions on the data which are saved in the databas

→ **static**

→ **dist**

- contains the supporting css for the UI

→ **js_scripts**

- contains the java scripts code that support the interaction of the user with the UI

→ **templates**

- contains the html files used for rendering the UI
  - the *html files starting with "start"* are used to create the Log-in page for each app
  - *query_teamplate.html* is used for rendering the Query Text Field
  - *task_description_template.html* is used for rendering the Task Description Field
  - *doc_ranking_display_information_template.html* is used for rendering the list of Display Fields for the UI Interaction App
  - *doc_ranking_view_information_template.html* is used for rendering the View Field of the Interaction Annotation UI when a user clicks on the View Button
  - *doc_annotate_profile_template.html* is used for rendering the Display Field and Annotation Field of the Score Annotation UI
  - *doc_ranking_display_compare_template.html* is used for rendering the Display Field of the Ranking Comparison UI
  - the *html files starting with "index"* are used to merge together the Query Field, Task Description Field and the Display Field of each app
  - *form_template.html* is used for rendering the exit survey
  - *stop_experiment_template.html* is used for rendering the final page of the annotation task

Templates used to render the XAI part:

- *doc_ranking_display_information_template_XAI.html* is the main wrapper of the XAI part containing the ranking list and view xai button
- *xai_section_template.html* is the wrapper for the specific XAI types
- *doc_ranking_view_information_factual_XAI_template.html* used to display the factual XAI containing the per feature display of feature importance values, accompanying image and accompanying text
- *doc_ranking_view_information_counterfactual_XAI_template.html used to display the counterfactual XAI* containing the updated item profile and accompanying text
- *doc_ranking_view_information_text_image_XAI_template.html* used to display the simple text XAI, image XAI and the text+image XAI
- *form_template_task_questionnaire.html* is used for rendering the questionnaire

→ **src**

→ **data_readers**

- contains the base class used to read the dataset in the format required by AnnoRank

→ **evaluate**
- contains evaluation scripts for computing utility metrics, fairness metrics and inter-annotator agreement

→ **fairness_interventions**
- contains the wrapper classes for the supported fairness interventions

  → **modules**
  - contains the supporting scripts for running of the fairness interventions

→ **rankers**
- contains the wrapper classes for the RankLib library

  →**modules**
  - contains the supporting scripts for running of the RankLib library

→ **utils**
- utilitary functions

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